

Robust Tracking by Adaptive Multi-feature Association

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Abstract Existing feature selection based object tracking schemes mostly assume initial target object appearance to be a stable reference in all subsequent feature evaluations. They also rank features independently without considering possible redundancies in describing a target object against its surrounding background or occluding distractors. Both assumptions can contribute to model drift leading to complete tracking failure over time. To address these problems, we formulate a novel framework that (1) introduces two attributes to rank features over time in order to reduce redundancy and improve the discriminability of the chosen set, (2) maintains multiple, selectively adaptive feature reference models verified by image evidence on-the-fly in order to cope with long-term changes in target object appearance, and (3) tracks objects by adaptively classifying pixels into object foreground and the surrounding dynamic background. We compare our model to three established schemes proposed by Collins et al (2005), Avidan (2007) and Grabner et al (2008) and demonstrate the significant performance advantages of our model in several challenging indoor and outdoor scenes.

Keywords Tracking · Feature selection · Model drift

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1 Introduction

Visual object tracking is rooted in the ability to determine those parts of an image that correspond to an object of interest and to use those parts to estimate object state such as position, orientation and size. Often, the approach taken has been to model the object itself and to perform object association with image data, using geometric information (e.g. Curwen and Blake (1992); Blake et al (1995); Gavrilu and Davis (1995); Drummond and Cipolla (2001); Plaenkers and Fua (2002); Snidaro et al (2008)), photometric (e.g. Comaniciu et al (2000); Raja et al (1998); Lucas and Kanade (1981); Matthews et al (2004)) or a combination of both (e.g. Takala and Pietikainen (2007)). Conversely, the background may be modelled and removed leaving regions taken to correspond to the foreground (e.g. Stauffer and Grimson (1999, 2000)).

Considering the problem more generally, object tracking methodologies must cope with imaging fluctuations caused by illumination change, partial or complete occlusions and ever-changing backgrounds caused by moving cameras or distractors in a multi-object environment. Tracking in a cluttered scene under unstable lighting is hindered by the intrinsic instability and transience of features as a useful discriminator for the target object. Furthermore, the robustness of different types of features is highly context-dependent, with photometric and geometric environmental conditions inducing different complications. For example, colour is sensitive to changes in lighting whilst shape and texture may be drastically altered during pose transitions. Additionally, in cluttered scenes dynamic distractors can significantly affect the relevance of specific features over time, e.g. colour may perform adequately when a red target object is tracked against a non-red background

but shape may be more discriminative when the target moves into a red-coloured area. Consequently, in the absence of truly robust features, a successful tracker will not only take measures to alleviate the problems of model-drift but also to utilise the features most likely to be useful at any given time.

In recent years, the problem of tracking objects with changing appearance against a changing background has been addressed by treating tracking as a binary classification problem (Comaniciu et al (2003); Lin et al (2004); Collins et al (2005); Nguyen and Smeulders (2006); Grabner et al (2006); Avidan (2007); Liang et al (2007); Stolkin et al (2007); Yu et al (2008)). Classification has also been used to disambiguate difficult situations in crowded scenes where the close proximity of objects may confuse a tracker (Song et al (2008)). In particular, for coping with constantly varying foreground and background appearance, a feature selection approach has increasingly been employed as a means to selecting the most discriminative appearance characteristics for performing object association given the conditions at any specific time. In this vein, Collins et al (2005) proposed a framework involving the continuous ranking of individual feature types from a pool over the duration of a tracking task and the use of high-ranked features to generate classifiers for pixel classification in subsequent frames. Periodically, foreground and background pixel samples collected from an object-centred centre-surround bounding box are used to evaluate the discriminative power for each of the features using the variance ratio of their statistics to determine the N most discriminative features. Classifiers are then created for these feature types and used to generate confidence maps to which mean-shift (Comaniciu et al (2000)) processes are then applied and combined. This idea was extended further by Avidan (2007) using AdaBoost (Freund and Schapire (1996)) to generate and maintain an ensemble of weak classifiers which may implicitly weight different features in a pool rather than perform selection. The weak classifiers are combined into a strong classifier used for generating confidence maps. Samples are collected for each frame and used to generate new weak classifiers in an off-line manner, replacing the weakest classifiers for the current ensemble. Similarly, Grabner et al (2006) proposed an on-line boosting algorithm to maintain an ensemble. Liang et al (2007) modified the framework by Collins et al (2005) to use Bayes error rates instead of the variance ratio to deal with multimodal feature distributions. They also included a method to adapt estimates of object scale using a set of simple correlation templates to roughly estimate object boundaries.

A major problem with modelling changing appearance over time in unconstrained environments such as these is *model drift*. Localisation errors can introduce contaminants into the foreground and background samples used to rank features and generate classifiers which build up over time, eventually leading to tracking failure. In order to avoid this, Collins et al (2005) and Liang et al (2007) incorporate a static reference model of target object appearance. This involves combining pixel samples from the first frame of a tracking task with frame-specific pixel samples when ranking. For similar effect, Avidan (2007) proposed exempting replacement of the strongest classifier from the first frame of tracking. Grabner et al (2008) took the more sophisticated approach of employing a semi-supervised on-line boosting approach based on the SemiBoost algorithm (Mallapragada et al (2008)). This involves the use of a fixed “prior” classifier acting as a static reference model and a separate strong classifier. In each frame, the strong classifier is used to generate confidence maps and estimate object state, after which randomly-selected unlabelled image patches are extracted and classified using a combination of both the strong and prior classifiers. The classes and corresponding weights of the image patches are then used to update the strong classifier. This approach affords the tracker greater flexibility to adapt without allowing it to lose relevance; however, the prior classifier is trained on initial labelled examples and remains static throughout the tracking process.

In this paper, we address two fundamental limitations of all of these previous methods. Firstly, the feature ranking schemes employed by Collins et al (2005) (using variance ratio) and Liang et al (2007) (using Bayes error) rank features in isolation which can result in highly redundant feature sets, giving significantly less discriminative representations of target objects. Boosting approaches, such as those used by Avidan (2007) and Grabner and Bischof (2006); Grabner et al (2006, 2008), can also suffer from the same problem (Fleuret (2004)). Secondly, the attempts at addressing model drift in Collins et al (2005); Liang et al (2007); Avidan (2007); Grabner et al (2008) all inherently either directly or indirectly assume the long-term relevance of target object appearance at initialisation, either by incorporating samples directly from the initial frame or using classifiers trained on a static set of initial samples. While these approaches may be effective in relatively stable environments where very short-term changes may take place, in situations where target object appearance undergoes long-term change, for example due to lighting, the maintained reference samples or classifier quickly become irrelevant. Both limitations contribute to inaccurate pixel classification which in turn results in

lower quality pixel confidence maps and more tracking errors.

To address these problems, we argue that combining knowledge of trends in target object appearance over time with object appearance on a frame-specific basis is a more natural approach to overcoming the problems caused by both short-term complications such as occlusion as well as longer-term issues such as lighting change. A key factor, and one neglected by previous methods, is that the feature selection approach to tracking requires a balance between rigidity and flexibility in feature selection. Too much rigidity and the tracker cannot adapt to new situations, too much flexibility and it becomes too susceptible to noise and model drift. Trends should be monitored over time to facilitate *dynamic* reference models as opposed to static. Doing so adds a degree of flexibility to the overall strategy for the tracking task and in the process improves the strength and stability of features selected as well as the accuracy of pixel classification from frame to frame. To that end, we propose a novel on-line adaptive feature ranking and selection framework that balances the long-term featural characteristics of an object with short-term discriminative requirements throughout tracking. We call this framework *Adaptive Multi-feature Association* (AMA) which comprises two components. Firstly, we perform object feature selection by ranking each object feature on the basis of two attributes in order to reduce redundancy in the chosen set: (a) discriminability; and (b) independence to other features. We call this *Attribute-based Feature Ranking* (AFR). Secondly, we maintain multiple feature reference models, one for each feature type in a pool and these are selectively adapted over time. We refer to these as *Multiple Selectively-adaptive Feature Models* (MSFM) which are combined with densities of current image data for object feature ranking and foreground pixel classification.

We compare the performance of our approach against the framework proposed by Collins et al (2005), Ensemble Tracking (Avidan (2007)) and the SemiBoost tracker (Grabner et al (2008)) in challenging outdoor and indoor scenarios and demonstrate that our scheme results in more stable feature selection, more accurate pixel classification and significantly more robust tracking under very difficult lighting and viewing conditions.

The paper is organised like so: in Section 2 we describe our framework and the corresponding algorithm. In Section 3 we describe a detailed analysis of six experiments conducted on four scenarios depicting brightness changes, colour changes, occlusion and combinations of these in difficult cluttered environments with low-resolution target objects at a distance from the camera. Finally, in Section 4 we offer conclusions.

2 Adaptive Multi-feature Association

We define feature selection and pixel classification based on a combination of current evidence and adaptive feature reference models for each feature type. More specifically, in each image frame t and for each feature type X_j , $j = 1..J$ from a feature pool of J features, pixels for foreground F_t and background B_t are gathered from an object centered centre-surround bounding box and corresponding densities $p(X_j|F_t)$ and $p(X_j|B_t)$ are generated. We employ a feature pool comprising linear combinations of RGB values (more details are given in Section 3). A set of J feature reference models $p(X_j|M_t)$, one for each feature type, are generated and maintained adaptively over time. The foreground density $p(X_j|F_t)$ at time t is combined with the corresponding feature reference model $p(X_j|M_t)$ to form a model describing a target object $p(X_j|O_t)$. Given this and $p(X_j|B_t)$, a set of classifiers are generated and ranked for each feature type and the N highest ranked are individually used for classifying pixels in the subsequent frame. For each of the N feature types, the confidence measures of those pixels classed as target object are used to generate confidence maps which are normalised. A mean-shift process is applied to the weighted average of the N maps to estimate target object position. The corresponding N feature reference models are then updated on the basis of previously observed foreground and background feature characteristics. This framework, which we call Adaptive Multi-feature Association (AMA), consists of two key components as follows: (1) Attribute-based Feature Ranking for selecting feature types¹ with reduced redundancy; and (2) Multiple Selectively-adaptive Feature Models for maintaining a long-term appearance reference for a tracked object. We now describe each of these in turn.

2.1 Attribute-based Feature Ranking

Feature selection for foreground pixel classification can be considered as choosing the K features $\{X_{b_1}, \dots, X_{b_K}\}$ from a pool of J features $X_1, \dots, X_J$ that minimise the conditional entropy $H(Y|X_{b_1}, \dots, X_{b_K})$, where Y is a class variable. Ideally, such a set is as small as possible to reduce descriptive redundancy amongst subsets of such features. However, this is in general computationally intractable. Algorithms exist that attempt to recover the Markov blanket of the class variable (e.g Yaramakala (2004) and more recently Fu and Desmarais (2008)); however, these are non-greedy algorithms which

¹ In the remainder of this paper, the term “features” refers to “feature types”.

can require many estimates of conditional independence to be made at each iteration, increasing computation cost which is undesirable for a time-critical task such as object tracking. The Conditional Mutual Information Maximisation (CMIM) algorithm (Fleuret (2004)) is a fast, principled suboptimal greedy method for finding such features wherein each feature selected maximises the mutual information with the class variable conditional on the features previously selected. However, this algorithm at each iteration requires the estimation of joint densities for three random variables, the class variable Y and two features being compared. Tracking small objects in low resolution images typically lacks sufficient data from each frame for a reliable estimate of such joint densities, which may make the use of such algorithms unreliable.

To address the problem of reducing redundancy, we approximate the choosing of descriptive features by computing two attributes per feature X_j for each frame t which we then average for ranking:

(1) **Discriminability** $d_{j,t}$ – Given K sample pixel feature values $\mathbf{x}_j^k$ for feature X_j from frame t , $k = 1..K$, $\mathbf{x}_j^k \in \mathcal{F}_j$ where $\mathcal{F}_j$ is the set of all possible values for feature X_j , we generate densities $p(X_j|F_t)$ and $p(X_j|B_t)$ for foreground and background respectively and compute the variance ratio as per Collins et al (2005):

$$d_{j,t} = \frac{\text{var} \{j, t, \frac{1}{2} [p(\cdot|F_t) + p(\cdot|B_t)]\}}{\text{var} \{j, t, p(\cdot|F_t)\} + \text{var} \{j, t, p(\cdot|B_t)\}} \quad (1)$$

where

$$\begin{aligned} \text{var} \{j, t, \Psi(\cdot)\} = & \left\{ \sum_{\mathbf{x} \in \mathcal{F}_j} \Psi(\mathbf{x}) \log^2 \left(\frac{p(\mathbf{x}|F_t)}{p(\mathbf{x}|B_t)} \right) \right\} \\ & - \left\{ \sum_{\mathbf{x} \in \mathcal{F}_j} \Psi(\mathbf{x}) \log \left(\frac{p(\mathbf{x}|F_t)}{p(\mathbf{x}|B_t)} \right) \right\}^2 \end{aligned} \quad (2)$$

The variance ratio could be replaced by another measure of discriminability such as classification error rate on the samples taken from frame t ; however, we use this to more clearly observe the influence of the second attribute (described below) on feature ranking as compared to Collins et al (2005).

(2) **Independence** $u_{j,t}$ – We approximate the degree of independence between features by computing one minus the average canonical correlation (Hotelling (1936)) between the K pixel samples collected for feature j , $\mathbf{x}_j^k \in R^m$ and the corresponding K pixel samples for all other features i from the feature pool, $\mathbf{x}_i^k \in R^n$, $i \neq j$.

We deploy Canonical Correlation Analysis (CCA) to find the basis functions $\langle \mathbf{w}_j, \mathbf{w}_i \rangle$ for which the projections $x_j = \mathbf{w}_j^T \mathbf{x}_j$ and $x_i = \mathbf{w}_i^T \mathbf{x}_i$ are maximally correlated. More specifically, CCA maximises the following:

$$\rho_{j,i} = \frac{E[x_j x_i]}{\sqrt{E[x_j^2]E[x_i^2]}} = \frac{E[\mathbf{w}_j^T \mathbf{x}_j \mathbf{x}_i^T \mathbf{w}_i]}{\sqrt{E[\mathbf{w}_j^T \mathbf{x}_j \mathbf{x}_j^T \mathbf{w}_j]E[\mathbf{w}_i^T \mathbf{x}_i \mathbf{x}_i^T \mathbf{w}_i]}} \quad (3)$$

A total of $V = \min(m, n)$ pairs of basis functions may be obtained $\langle \mathbf{w}_j^v, \mathbf{w}_i^v \rangle$, $v = 1..V$, with corresponding correlations $\rho_{j,i}^v$ by successively solving Equation 3 subject to the constraint that $\langle \mathbf{w}_j^v, \mathbf{w}_i^v \rangle$ are orthogonal to all preceding pairs found. We then estimate a measure of independence $u_{j,t}$ as the average over the V dimensions and correlations between feature j and all other features i in the pool:

$$u_{j,t} = 1 - \frac{1}{V(J-1)} \sum_{i \neq j} \sum_v \rho_{j,i}^v \quad (4)$$

The final feature ranking $r_{j,t}$ is computed as the sum of Discriminability and Independence:

$$r_{j,t} = d_{j,t} + u_{j,t} \quad (5)$$

By combining these two attributes, the rankings for features considered strongly discriminative by variance ratio are enhanced according to their level of correlation with other features in the pool. Highly-correlating features are considered as embodying redundant information and their ranking value $r_{j,t}$ increased less than those with low average correlation with the other features.

2.2 Multiple Selectively-adaptive Feature Models

Rather than maintain a static set of reference pixels for combining with current image data at each frame (Collins et al (2005); Liang et al (2007); Avidan (2007); Grabner et al (2008)), we consider multiple adaptive feature reference models $p(X_j|M_t)$, one for each feature X_j in the pool, which are defined in terms of previously observed foreground and background characteristics and adapted during tracking to maintain the most useful target object discriminators within each feature domain. Adaptation is performed only for the N highest ranked features in order to maintain a level of confidence in the data being used for the update. More precisely, the conditional density $p(X_j|M_t)$ for feature X_j at frame t is:

$$p(X_j|M_t) = \sum_{l=t-L+1}^t \{P_j(F_l)p(X_j|F_l) - P_j(B_l)p(X_j|B_l)\} \quad (6)$$

where L is a temporal window controlling the rate of adaptation of feature reference models, l indexes frames within the window, $p(X_j|F_l)$ and $p(X_j|B_l)$ are foreground and background densities respectively and corresponding priors are $P_j(F_l)$ and $P_j(B_l)$. A smaller value for L enables the model to adapt more strongly to shorter-term changes. With this formulation, a given feature vector is essentially weighted by its foreground representation minus its background representation over the temporal window. Clearly, over time, those feature vectors which are statistically stronger for the foreground reinforce their representation in the model in proportion to their strength, and similarly weakened ones are more associated with the background.

2.2.1 Estimating Priors

The prior probabilities $P_j(F_t)$ and $P_j(B_t)$ reflect *confidence* in the descriptive accuracy of the foreground and background sample distributions respectively for feature X_j in frame t . These are important for updating the feature reference models (Equation 6) to reduce the chance of model drift over time. In updating these priors, Bayes' error estimates may be used to derive two characteristics for distributions of foreground and background data; *stability*, indicating the degree of change in the distribution between frames, and *unreliability*, relating to an overlap between foreground and background. These are then combined to derive a final estimate for the prior. More precisely, the foreground and background priors $P_j(F_t)$ and $P_j(B_t)$ respectively for feature X_j at frame t are computed like so:

$$P_j(F_t) = \frac{1}{2} \left(1 + \int_{X_j} \min \{p(X_j|F_t), p(X_j|F_{t-1})\} dX_j - \int_{X_j} \min \{p(X_j|F_t), p(X_j|B_{t-1})\} dX_j \right) \quad (7)$$

$$P_j(B_t) = \frac{1}{2} \left(1 + \int_{X_j} \min \{p(X_j|B_t), p(X_j|B_{t-1})\} dX_j - \int_{X_j} \min \{p(X_j|B_t), p(X_j|F_{t-1})\} dX_j \right) \quad (8)$$

For each prior, the first integral represents the match between the current and previous densities for the corresponding class with high values indicating good stability. The second integral is proportional to the degree of overlap between the classes and indicates the level of unreliability of the corresponding class density.

2.2.2 Histogram Approximation

In practice, we approximate Equations 1, 2, 6, 7 and 8 by modelling foreground, background and feature reference distributions as histograms $h(\mathbf{x}_j|F_t)$, $h(\mathbf{x}_j|B_t)$ and $h(\mathbf{x}_j|M_t)$ with bins corresponding to discretised values for feature X_j . The histogram for the corresponding reference model is then updated in frame t by:

$$h(\mathbf{x}_j^k|M_t) = \sum_{l=t-L+1}^t \max \{0, [P_j(F_l)h(\mathbf{x}_j^k|F_l) - P_j(B_l)h(\mathbf{x}_j^k|B_l)]\} \quad (9)$$

for all bins k with corresponding feature vector value $\mathbf{x}_j^k$. The priors are estimated using histogram intersection:

$$P_j(F_t) = \frac{1}{2} \left(1 + \sum_k \min \{h(\mathbf{x}_j^k|F_t), h(\mathbf{x}_j^k|F_{t-1})\} - \sum_k \min \{h(\mathbf{x}_j^k|F_t), h(\mathbf{x}_j^k|B_{t-1})\} \right) \quad (10)$$

$$P_j(B_t) = \frac{1}{2} \left(1 + \sum_k \min \{h(\mathbf{x}_j^k|B_t), h(\mathbf{x}_j^k|B_{t-1})\} - \sum_k \min \{h(\mathbf{x}_j^k|B_t), h(\mathbf{x}_j^k|F_{t-1})\} \right) \quad (11)$$

The resulting conditional probabilities are derived from the normalised histograms:

$$\begin{aligned} P(\mathbf{x}_j^k|M_t) &= \frac{h(\mathbf{x}_j^k|M_t)}{\sum_i h(\mathbf{x}_j^i|M_t)} \\ P(\mathbf{x}_j^k|F_t) &= \frac{h(\mathbf{x}_j^k|F_t)}{\sum_i h(\mathbf{x}_j^i|F_t)} \\ P(\mathbf{x}_j^k|B_t) &= \frac{h(\mathbf{x}_j^k|B_t)}{\sum_i h(\mathbf{x}_j^i|B_t)} \end{aligned} \quad (12)$$

Note that the histograms are maintained in their unnormalised state and only normalised during the computation of the conditional probabilities. This is to ensure that the strength of representation for feature values in

the feature models $h(\mathbf{x}_j^k|M_t)$ are not capped during the update computation (Equation 9).

In summary, we maintain multiple dynamic reference models, one for each feature domain, which are selectively updated over time using Equation 6. These incorporate image evidence over the previous L frames, with the foreground and background priors $P_j(F_l)$ and $P_j(B_l)$ from each frame weighting the corresponding evidence. These priors are estimated according to Equations 7 and 8, each composed of two Bayes' error terms indicating *stability* and *unreliability*. The *stability* term is computed by comparing evidence from the current frame and the previous frame for the same class (e.g. between foreground data for frame t and foreground data from frame $t-1$), indicating the level of change between the distributions. A high Bayes' error indicates high stability since change is small. Conversely, a low Bayes' error indicates a sudden difference in the distributions which can be caused by tracker localisation error or occlusion, in which case stability is low. The *unreliability* term relates to the match between the statistics for different classes between frames (e.g. between foreground data from frame t and background data from frame t). A high Bayes' error here indicates a strong overlap between the classes and consequently high unreliability. Conversely, a low Bayes' error corresponds to low overlap and greater confidence in the data differentiating the two classes. Each prior is then constructed by subtracting the value for unreliability from the value for stability and scaling and shifting the result to ensure a value between zero and one. Low priors directly relate to the level of confidence in the corresponding image data, either as a result of tracking failure or occlusion, with their influence on the reference models consequently reduced.

2.3 Tracking by Adaptive Multi-feature Association

There are three steps for the overall tracking strategy. Firstly, attention is focused on a specific region of the image within which the object is expected to lie. Secondly, features are ranked according to estimates of their discriminability using samples from the previous frame. Finally, a confidence map is generated from a weighted combination of the most highly ranked features by classifying pixels as object or background within the region of focus-of-attention. This confidence map is used to estimate object position.

A target-centered approach is deployed to sample foreground and background pixels in each frame for the feature ranking step. The background is assumed to be subject to spontaneous and continuous change as would be expected for situations with moving cameras

or busy scenes with multiple moving occluding or proximal distractors. More specifically, an object-centred box of width w and height h bounds a tracked object. Pixels within are used as foreground feature samples and background appearance is sampled from a region of thickness $\beta \cdot \max(w, h)$ surrounding the centre box. The union of pixels from the centre and surround regions are denoted as region R .

In the first frame $t = 1$ of tracking, given an initialised centre box for a target object O in a scene, the priors for foreground and background $P_j(F_1)$ and $P_j(B_1)$ are set to 0.5 and foreground and background densities $p(X_j|F_1)$ and $p(X_j|B_1)$ generated from the expanded region R . Then the feature reference models $p(X_j|M_1)$ for each feature X_j are updated. The foreground and background densities are then used for Attribute-based Feature Ranking as described in Section 2.1. The best N features are used for the generation of confidence maps in the second frame, which are combined and used for mean-shift estimation of a new box position. In any given frame t , the confidence map $c_{j,t}$ for feature X_j is generated using models from the previous frame by computing the Bayesian posterior $P(O_t|\mathbf{x}_j^k)$ for the feature vector values $\mathbf{x}_j^k$ for all K pixels $k \in R_{t-1}$ of the centre-surround region R_{t-1} estimated for the previous frame:

$$P(O_t|\mathbf{x}_j^k) = \frac{P(\mathbf{x}_j^k|O_{t-1})P_j(O_{t-1})}{P_{t-1}(\mathbf{x}_j^k)} \quad (13)$$

where

$$P_{t-1}(\mathbf{x}_j^k) = P(\mathbf{x}_j^k|O_{t-1})P_j(O_{t-1}) + P(\mathbf{x}_j^k|B_{t-1})P_j(B_{t-1}) \quad (14)$$

and where the priors for object O_{t-1} and background B_{t-1} sum to unity, $P_j(O_{t-1}) + P_j(B_{t-1}) = 1$ and the conditional probability $P(\mathbf{x}_j^k|O_{t-1})$ is computed as:

$$P(\mathbf{x}_j^k|O_{t-1}) = P(\mathbf{x}_j^k|F_{t-1})P_j(F_{t-1}) + P(\mathbf{x}_j^k|M_{t-1})P_j(M_{t-1}) \quad (15)$$

Here we have two new priors, $P_j(M_{t-1})$ and $P_j(O_{t-1})$ for the reference models and object respectively. These can be computed simply as:

$$P_j(M_{t-1}) = 1 - P_j(F_{t-1}) \quad (16)$$

$$P_j(O_{t-1}) = 1 - P_j(B_{t-1}) \quad (17)$$

from the foreground and background priors for frame $t-1$ (Equations 7 and 8). A high value for $P_j(F_{t-1})$ results in less emphasis being placed on the reference model

M_{t-1} and vice-versa (Equation 15). Consequently, low confidence in current image evidence results in more weight being placed on the reference model.

In practice we use the log-likelihood ratio to classify pixels with the sign denoting class and magnitude the confidence. For the top N features' confidence maps $c_{n,t}(k)$, $n = 1..N$ where k indexes a pixel, we set all negative (background) values to zero and derive the final confidence map $C_t(k)$ as a weighted average of the normalised individual maps:

$$c_{n,t}(k) = \max \left\{ 0, \log \left(\frac{P(\mathbf{x}_n^k | O_{t-1}) P_n(O_{t-1})}{P(\mathbf{x}_n^k | B_{t-1}) P_n(B_{t-1})} \right) \right\} \quad (18)$$

$$C_t(k) = \sum_n \frac{r_{n,t-1}}{\sum_{i=1}^N r_{i,t-1}} \cdot \frac{c_{n,t}(k)}{\sum_{l=1}^K c_{n,t}(l)} \quad (19)$$

for the K pixels $k \in R_{t-1}$ where $r_{n,t-1}$ is the ranking score for feature n (Equation 5).

Mean-shift is then applied to $C_t(k)$ to estimate the new target object position and the corresponding N feature reference models updated (Equation 9). The new centre-surround box is used to separate new foreground and background samples for the cycle to continue in subsequent frames.

In summary, given an initialised bounding box for the first frame, the bounding box is expanded to yield a surround region and foreground and background distributions $P(X_j|F_1)$ and $P(X_j|B_1)$ generated accordingly from this expanded region. All priors $P_j(F_1)$, $P_j(B_1)$, $P_j(M_1)$ and $P_j(O_1)$ are set to 0.5 and all feature reference models $P(X_j|M_1)$ updated as per Equation 6. This then initiates the main tracking cycle which involves:

1. Ranking each feature X_j according to the current foreground and background sample data $P(X_j|F_t)$ and $P(X_j|B_t)$;
2. Loading the next frame $t + 1$;
3. Computing a confidence map for the new frame given the N highest ranked features;
4. Estimating a new bounding box given the current confidence map;
5. Scaling the new bounding box to yield a centre-surround region;
6. Collecting data for new foreground and background distributions $P(X_j|F_{t+1})$ and $P(X_j|B_{t+1})$;
7. Updating the priors $P_j(F_{t+1})$, $P_j(B_{t+1})$, $P_j(M_{t+1})$ and $P_j(O_{t+1})$; and
8. Updating the reference models $P(X_j|M_{t+1})$ for the top N ranked features.

The cycle then repeats. This procedure is detailed in Algorithm 1.

Input: Sequence of frames $f_1, \dots, f_T$, object bounding box b_1 for frame f_1
Output: Bounding boxes $b_2, \dots, b_T$ for frames $f_2, \dots, f_T$
 Initialise all feature reference models $p(X_j|M_1)$ to null;
 Set bounding box $v = b_1$ with width w and height h ;
foreach $t = 2$ **to** T **do**
 Expand v by $\beta \cdot \max(w, h)$ for surround region;
 Generate $p(X_j|F_t)$ and $p(X_j|B_t)$ for each feature X_j ;
 if $t = 2$ **then**
 Set $P_j(F_1) = P_j(B_1) = P_j(M_1) = P_j(O_1) = 0.5$;
 Update $p(X_j|M_1)$ for all features using Equations 9 and 12;
 else
 Update $P_j(F_{t-1})$, $P_j(B_{t-1})$, $P_j(M_{t-1})$ and $P_j(O_{t-1})$ using Equations 7, 8, 16 and 17;
 Update $p(X_j|M_{t-1})$ for top N ranked features using Equations 9 and 12;
 end
 Rank each feature X_j using Equations 1, 4 and 5;
 Get frame f_t ;
 Generate confidence map C_t for f_t using Equation 18;
 Apply mean-shift to C_t for new target position estimate and bounding box b_t ;
 Set $v = b_t$;
end

Algorithm 1: The Adaptive Multi-feature Association (AMA) algorithm. Tracking initialisation is assumed to be provided along with the first frame. All priors are initialised to 0.5 and these values used for the second frame; they are updated for subsequent frames using Equations 7, 8, 16 and 17. Reference models are then updated using Equations 9 and 12 (all features if the first frame, only the top N if subsequent). All features are then ranked using Equations 1, 4 and 5. The next frame is then collected and confidence maps computed using the log-likelihood ratio (Equation 18). Finally, mean-shift is applied to estimate new object position and the cycle restarts.

3 Experiments

We conducted experiments on four scenarios labelled A, B, C and D for comparing the following; (a) the framework presented by Collins et al (2005); (b) Ensemble Tracking (Avidan (2007)); (c) The SemiBoost tracker (Grabner et al (2008)); and (d) our Adaptive Multi-feature Association (AMA) method. For (d), we show results both for: (1) Attribute-based Feature Ranking (AFR) only, i.e. with static feature reference models in the vein of previous methods; and (2) AFR combined with our Multiple Selectively-adapted Feature Models (MSFM) framework to illustrate how the latter further improves robustness.

3.1 Datasets and Settings

Our scenarios consisted of one indoor scene from a course of a train station and three outdoor scenes from the PETS 2009 dataset (of Reading (2009)). They are all wide-angle scenes containing multiple moving dis-

tractors and low-resolution objects undergoing temporary occlusions. Scenario A is taken from the PETS 2009 database and consists of 40 frames of a crowded scene with complex shadowed and bright regions. This is used to compare the strength and stability of the highest ranked features for each of the methods along with the quality of the resulting confidence maps. Scenario B consists of 250 frames from a low-quality scene of a concourse at a train station and is used to demonstrate the greater robustness and flexibility of AMA in tracking during severe lighting changes. Scenario C is also from PETS 2009 and consists of around 85 frames of a crowd of people moving against high-contrast bright and shadowed regions. This is used to demonstrate the ability of AMA to cope with the temporary severe occlusion of a small target object by other moving objects. Finally, Scenario D is another sequence from PETS 2009 consisting of 160 frames showing a group of people moving to congregate as a crowd in the centre of the scene. In this scenario we demonstrate the robustness of AMA in tracking a small target object undergoing both severe occlusion and severe lighting changes simultaneously. In each scenario, we track a single person as they move through the scene.

3.2 Implementations

All implementations were done in MATLAB. For all trackers, we employed the same feature pool as Collins et al (2005) which comprised 49 unique linear combinations of R, G and B values with possible multipliers $\{-2, -1, 0, 1, 2\}$. Values were scaled to between 0 and 31 and densities modelled as normalised 32-bin histograms.

We implemented the Collins et al (2005) tracker framework largely as described in the original paper; however, since our experiments focus on feature selection and model drift, we did not implement their technique for disambiguating distractors. As per the original work, we retained samples from the first frame (effectively a static reference model) which we combined with evidence in individual frames in order to alleviate model drift. For Ensemble Tracking (Avidan (2007)) we employed the weak classifiers used in the original paper, with the data comprising 49-dimensional feature vectors per pixel, each value representing one of the features from the feature pool. Confidence maps were generated from the dot product between a separating hyperplane and the feature vectors; consequently, the hyperplane acted as a weighting for each of the features in the pool. The strong classifier consisted of five such weak classifiers, with the result that different combinations of weighted features from the pool were used over

time. In each frame, the single weakest classifier was replaced by a newly computed one from new sample data. The strongest classifier from the first frame was retained throughout tracking, acting as a static reference model to counteract model drift. The SemiBoost tracker was implemented again according to the method and algorithm given in Grabner et al (2008). We maintained a strong classifier comprising the strongest weak classifiers from three selectors, with each selector containing 49 weak classifiers, one for each feature from the pool. We employed the same classifiers used for the Collins et al. method as well as for AMA. As per Grabner et al (2008), a strong classifier, trained on the first frame, acted as a prior (i.e. a static reference model) to reduce model drift. In all cases, we employed a mean-shift operator to update estimates of object position in each frame from the confidence maps generated by each tracker.

For our AMA implementation, we set the region multiplier for the background β (see Section 2.3) to 0.75. The window for updating feature reference models w was set to 50, i.e. sufficiently long (Section 2.2). We used the top $N = 3$ ranked features for confidence map generation and model updates, since using more was previously reported to offer little extra for this feature pool (Collins et al (2005)). When testing AFR only, we used static feature reference models initialised in the first frame as per Collins et al (2005) and Liang et al (2007). Accordingly, we also set all priors for each feature j , $P_j(O)$, $P_j(B)$, $P_j(F)$ and $P_j(M)$ (Section 2.1) to a constant 0.5. When testing AFR in combination with MSFM, the priors were updated as per Equations 7, 8, 16 and 17.

We now describe in turn the experiments conducted on all four scenarios as described above. For each scenario and for Collins et al., Ensemble Tracking, the SemiBoost tracker and AMA (both AFR only and AFR + MSFM), we show: (a) several output frames with overlaid estimated bounding boxes together with manually labelled ground truth boxes to illustrate object localisation; (b) corresponding pixel confidence maps to show the accuracy of pixel classification; (c) statistics of highest ranked features during tracking to illustrate the level of stability and consistency in feature association; and (d) plots illustrating localisation errors in terms of pixel displacement between estimated bounding boxes and manually-labelled ground truth.

3.3 Scenario A: Feature Selection Quality

The feature selection approach to tracking requires a balance between rigidity and flexibility in feature selection. Too much rigidity and the tracker cannot adapt

to new situations, too much flexibility and it becomes too susceptible to noise and model drift. These problems are largely reflected in the quality of the resulting confidence maps from which object positions are estimated. Cleaner and sparser maps are more desirable since they provide fewer distractors for estimating object position. In this experiment, we compare feature ranking statistics and generated confidence maps for each of the trackers.

Figure 1 shows the output from each of the trackers for frames 10, 20, 30, 40 and 50 of Scenario A. Each row (a)-(e) illustrates the application of Collins et al., Ensemble Tracking (ET), the SemiBoost tracker (ST), AFR and AFR+MSFM respectively. Each frame shows the output from the tracker with estimated object centre-surround bounding boxes (small yellow and large red boxes respectively). Figure 2 shows the corresponding confidence maps generated for the top ranked feature. Although for this example all the trackers maintained a lock on the object successfully (as confirmed by the ground-truth comparison in Figure 4), note that with AFR the part of the target most salient in discriminating against distractors was evidently emphasised in the confidence maps (the person's blue top) as compared with Collins et al. and Ensemble Tracking, demonstrating the greater utility of the features selected. The SemiBoost tracker settled to even cleaner confidence maps after Frame 20. However, the maps for AFR+MSFM were consistently the sparsest and most object specific from the beginning and throughout the duration of the sequence. This is attributed to both a better initial selection of features as well as a tendency of the adaptive models to strengthen the representation of consistently observed feature values over time, which in turn facilitated more stability in future feature selection and accurate pixel classification.

Figure 3 shows plots corresponding to the top three ranked (in the case of Ensemble Tracking and the SemiBoost tracker, weighted) features for each frame of tracking from Scenario A. It can be seen that Collins et al. and Ensemble Tracking showed greater fluctuation and instability in their selection of features. This is reflected in the more noisy confidence maps seen in Figures 2(a) and 2(b). However, AFR was able to overall select more stable and appropriate features during the tracking task. Using MSFM further stabilised the selection process and improved pixel classification. This was reflected in the tighter and cleaner confidence maps from Figures 2(d) and 2(e).

Although the SemiBoost tracker showed the least variation in highest ranked features, the corresponding confidence maps were not consistently sparse throughout the sequence and showed more noise early on (Fig-

ure 2(c)), suggesting that those ranked most highly at the beginning were not necessarily the most useful. The feature ranking statistics for the SemiBoost tracker (Figure 3) also showed the single same feature being ranked highest from around frame 20 onwards, with the next two best features essentially being discarded by having their weights set to zero. This coincides with the cleaner confidence maps shown in Figure 2(c). This turned out to be acceptable for this scenario as the tracker was able to perform successfully; however, as will be shown in the following scenarios it suggests a lack of flexibility which can be detrimental in more challenging tracking situations.

3.4 Scenario B: Tracking under Severe Illumination Changes

We now examine tracker performance for a challenging indoor scenario comprising 250 frames undergoing severe lighting change over time. We have two examples of changing illuminant for this scenario; the first showing a change in brightness and the second a change in colour. These changes are simulated by gradually modifying the original pixel values in sequence frames over time. For brightness changes, these modifications are then gradually reversed to original conditions. This allows us to observe the level of robustness of each of the trackers as well as their ability to adapt to such changing conditions.

3.4.1 Example 1 - Brightness Change

Figure 5 shows key output frames from each tracker when applied to the sequence undergoing brightness change. Brightness was gradually reduced over a period of around 125 frames (around four seconds in real time) before gradually being increased at the same rate to original conditions. Specifically, Figures 5 (a)-(e) show tracking outputs for Collins et al., Ensemble Tracking (ET), the SemiBoost tracker (ST), AFR and AFR + MSFM respectively. Figure 6 shows corresponding confidence maps generated from a weighted combination of the top three ranked features for each tracker. The SemiBoost tracker was distracted very early on, as shown in the second column. Collins et al. and Ensemble Tracking eventually failed by Frame 5867 whereas AFR and AFR + MSFM were able to maintain a lock on the target object.

The AFR tracker failed in Frame 5876, by which time AFR + MSFM was seemingly consistently displaced from the target. Interestingly however, as can be seen by a further examination in Figure 9 and the final column of Figure 5, this proved to be a precursor to

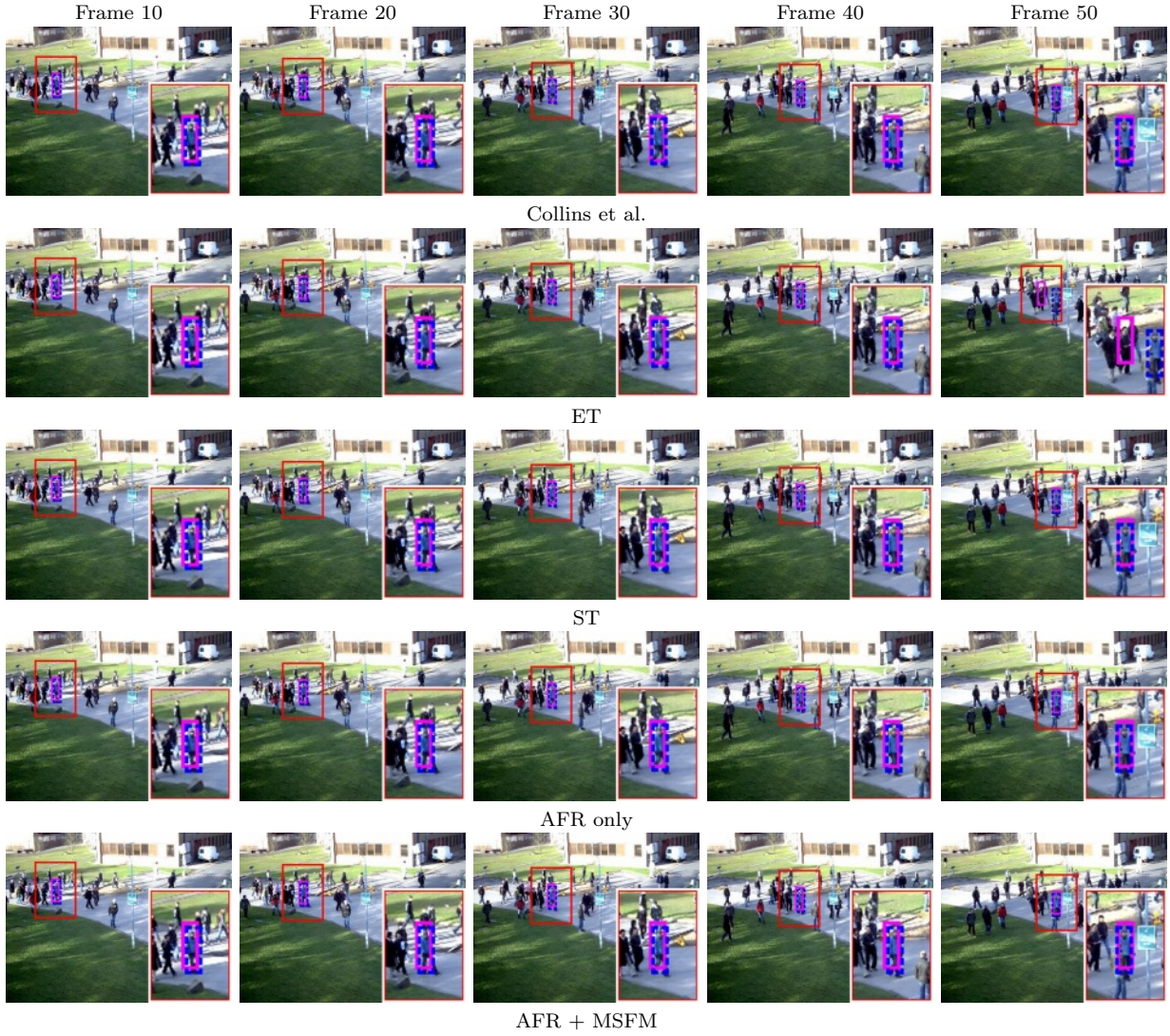


Fig. 1 Scenario A Tracker Output: The centre-surround box shows tracker localisation for the object, with the center box denoting estimated object position and the surround region the area from which background samples are gathered. The dashed blue box indicates manually labelled ground truth. The bottom-left of each output frame depicts the tracked region zoomed in for clarity. See Figure 2 for corresponding confidence maps.

tracker recovery. From just after Frame 5860, the lighting became dark enough for the tracker to no longer be able to distinguish the target; however, as can be seen in Frames 5880, 5900 and 5920 of Figure 9(a), the tracker appeared to attach itself to the bag being carried by the person. This was reflected by the slight offset from the center of the person as shown in the ground truth plot in Figure 9(b). The plot in Figure 9(c) shows the dominant feature for frames from 5850 onwards. It can be seen that up to the point when the target became too dark, feature number 16 was consistently the strongest feature. During this period, its corresponding feature reference model was continually updated. When the tracker lost its target, different features became dominant in

turn. At this point, the reference model for feature 16 was effectively frozen since it was no longer relevant and reliable samples for update could not be collected. A different set of dominant features emerged with feature 12 particularly relevant to tracking the bag belonging to the person, up to just after Frame 5950 when the person became salient enough to be recaptured. At this point, Feature 16 was once again consistently ranked the most stable and dominant, whereupon its reference model became active again and the tracker resumed where it left off prior to its temporary failure. It is interesting to note that, due to the period of dominance of feature 12, the representation of the bag was strengthened during the darker period of the sequence via the



Fig. 2 Scenario A Confidence Maps: The confidence maps correspond with the bounding box regions shown in the frames of Figure 1. AFR shows cleaner, sparser confidence maps than both Collins et al. and Ensemble Tracking (ET) where the most salient part of the object is more emphasised. The SemiBoost tracker (ST) shows sparser maps after frame 20. However, adding MSFM to AFR improves the AFR maps and further increases emphasis on the salient object region consistently for all frames, demonstrating the greater specificity of the adaptive models for AFR+MSFM.

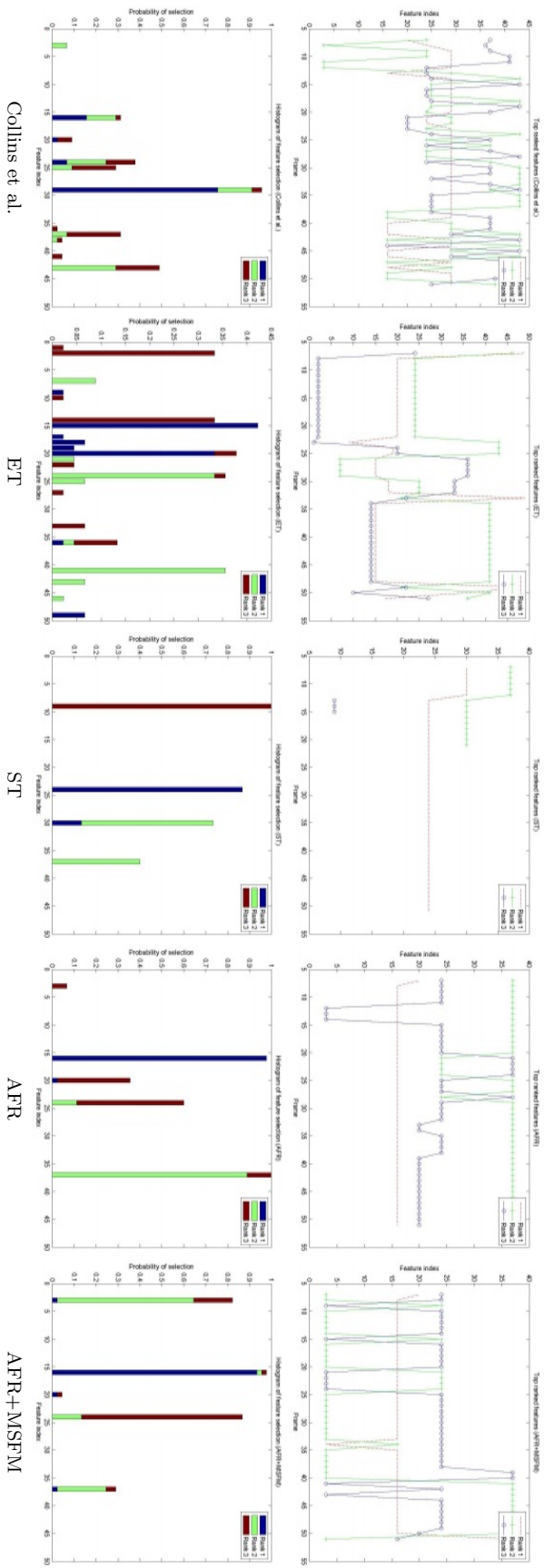


Fig. 3 Scenario A feature selection statistics. Top row shows top three ranked (or weighted) features for each frame and the second row the corresponding frequencies of selection for each feature in the pool. Only features with non-zero weights are included. The features selected for AFR and AFR+MSFM were better discriminators and prone to fewer fluctuations than both Collins et al. and Ensemble Tracking (ET). This resulted in cleaner confidence maps (see Figure 2). The SemiBoost tracker (ST) showed the least variation in feature selection. From frame 20 onwards, the same single feature was effectively selected for classification, with all other features weighted zero.

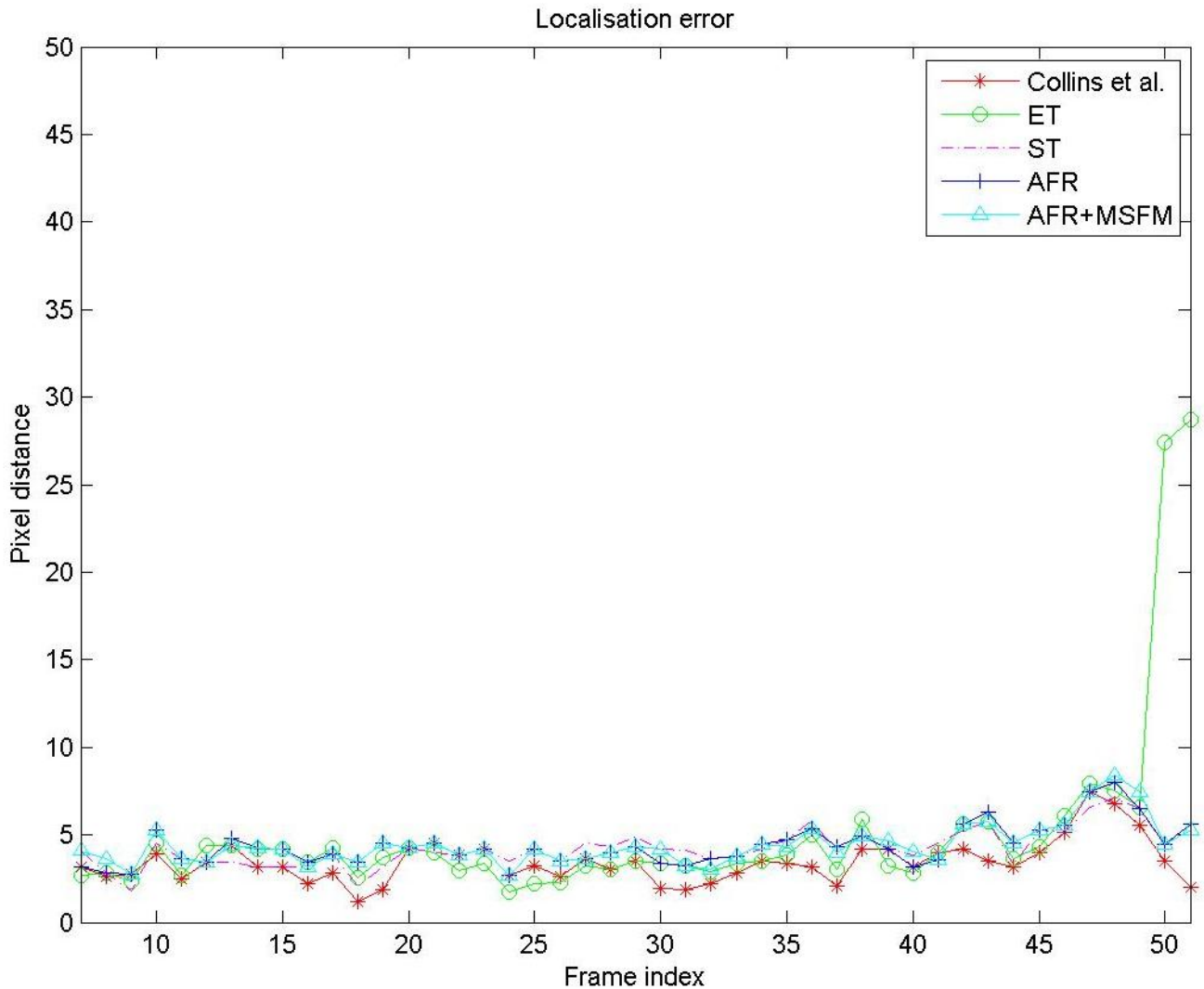


Fig. 4 Scenario A localisation errors in pixel distance between each tracker and manually-labelled ground truth. All trackers performed well for this scenario, with only Ensemble Tracking deviating in the last couple of frames.

adaptive reference models and consequently, following reacquisition of the person, the bag itself became an integral component of the representation for tracking, as exemplified by the fluctuation in highest rank between features 16 and 12 (for person and bag respectively) from frame 5950 onwards. This illustrates the advantages of adaptive reference models in maintaining the association between object and features in a dynamic way under changing conditions.

As can be seen from the confidence maps in Figure 6, the most salient part of the target object (the red coat) was strongly reflected since AFR ranked the more appropriate features more highly (while the tracker was still attached to the target). Using MSFM in addition further improved the confidence maps since the feature reference models were updated to reflect the more

strongly discriminative feature values, which in turn contributed more to greater classification accuracy.

Figure 7 shows the feature selection statistics for the top three ranked features for this example. Here, compared to Collins et al. and Ensemble Tracking (ET), we see a less haphazard ranking of features when using AFR or AFR+MSFM. The methods resulted in tighter, more representative confidence maps and more robust tracking as seen in Figure 5 (d) and (e). The SemiBoost tracker (ST) had the most rigid record of feature ranking; however, early failure of the tracker renders this moot and suggests that this seemingly less haphazard ranking behaviour is not related to the selection of the most appropriate features in practice. Figure 8 gives an overview of tracking performance by plotting tracker

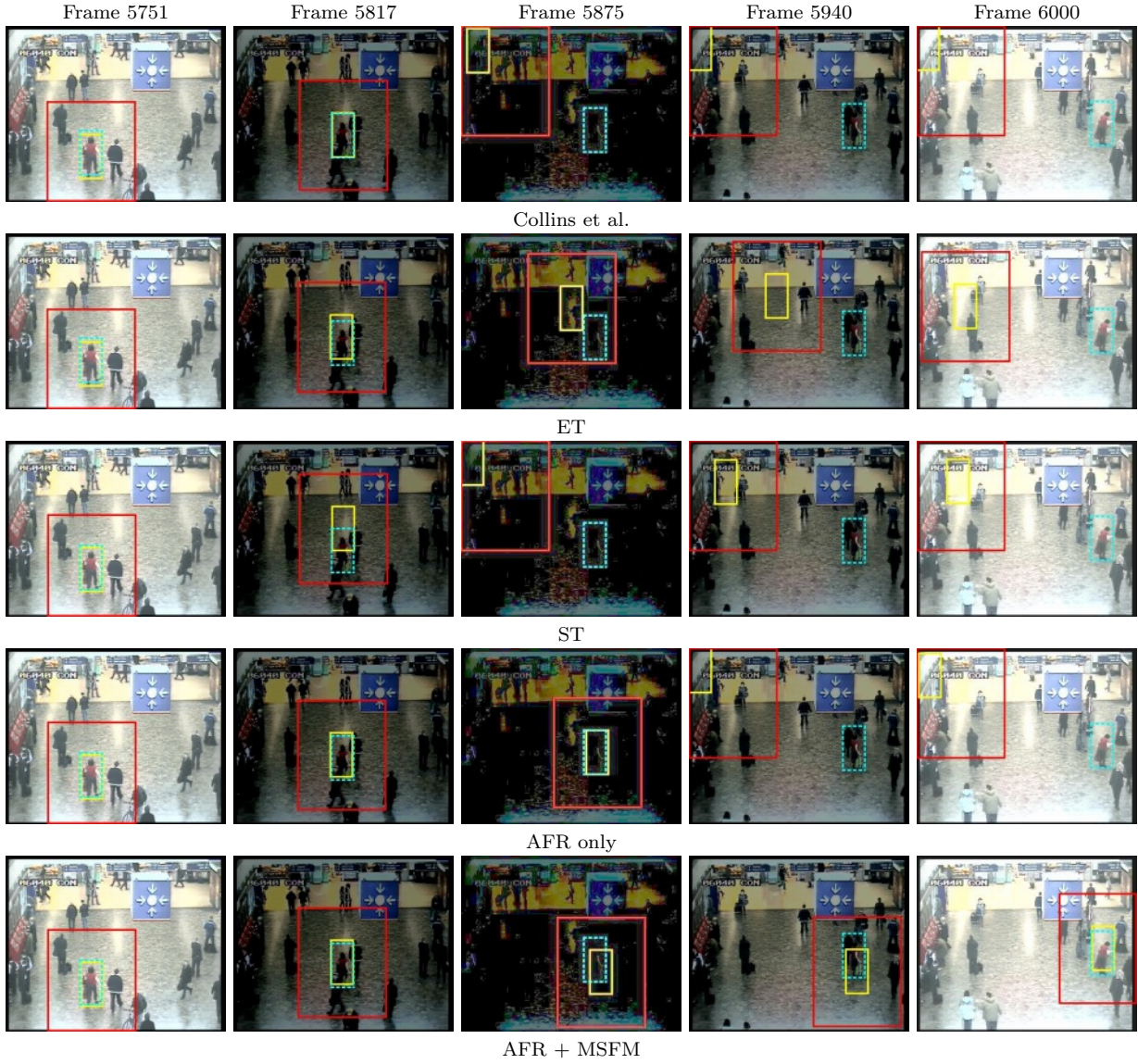


Fig. 5 Scenario B, Example 1 Tracker Output: Severe brightness change. Frame 5875 is gamma corrected for reader clarity. Brightness was reduced until around halfway through the sequence when it was increased again to the original conditions. The centre-surround box shows tracker localisation for the object. The dashed blue box indicates manually labelled ground truth. Collins et al., Ensemble Tracking (ET) and the SemiBoost tracker (ST) all failed to maintain tracking. AFR managed to track for longer but failed when the scene became extremely dark. AFR+MSFM was able to adapt to track the person's bag which was still sufficiently salient, until the person became strongly visible enough to be recaptured by the tracker for the remainder of the sequence. See Figure 6 for corresponding confidence maps.

accuracy as pixel displacements from manually-labelled ground truth.

3.4.2 Example 2 - Colour Change

Figure 10 shows some output frames from each tracker when applied to the sequence undergoing colour change. The red channel was gradually reduced and the blue channel increased over a period of around 125 frames (around four seconds in real time). Specifically, Fig-

ures 10 (a)-(e) show tracking outputs for the trackers of Collins et al., Ensemble Tracking (ET), the SemiBoost tracker (ST), AFR and AFR+MSFM respectively. Figure 11 shows the corresponding confidence maps. Ensemble Tracking and the SemiBoost tracker were distracted prior to halfway through the sequence whereas Collins et al. became attached to another moving person later on. Both AFR and AFR+MSFM were able to maintain a lock on the target object for the duration of the sequence. This is further illustrated in the

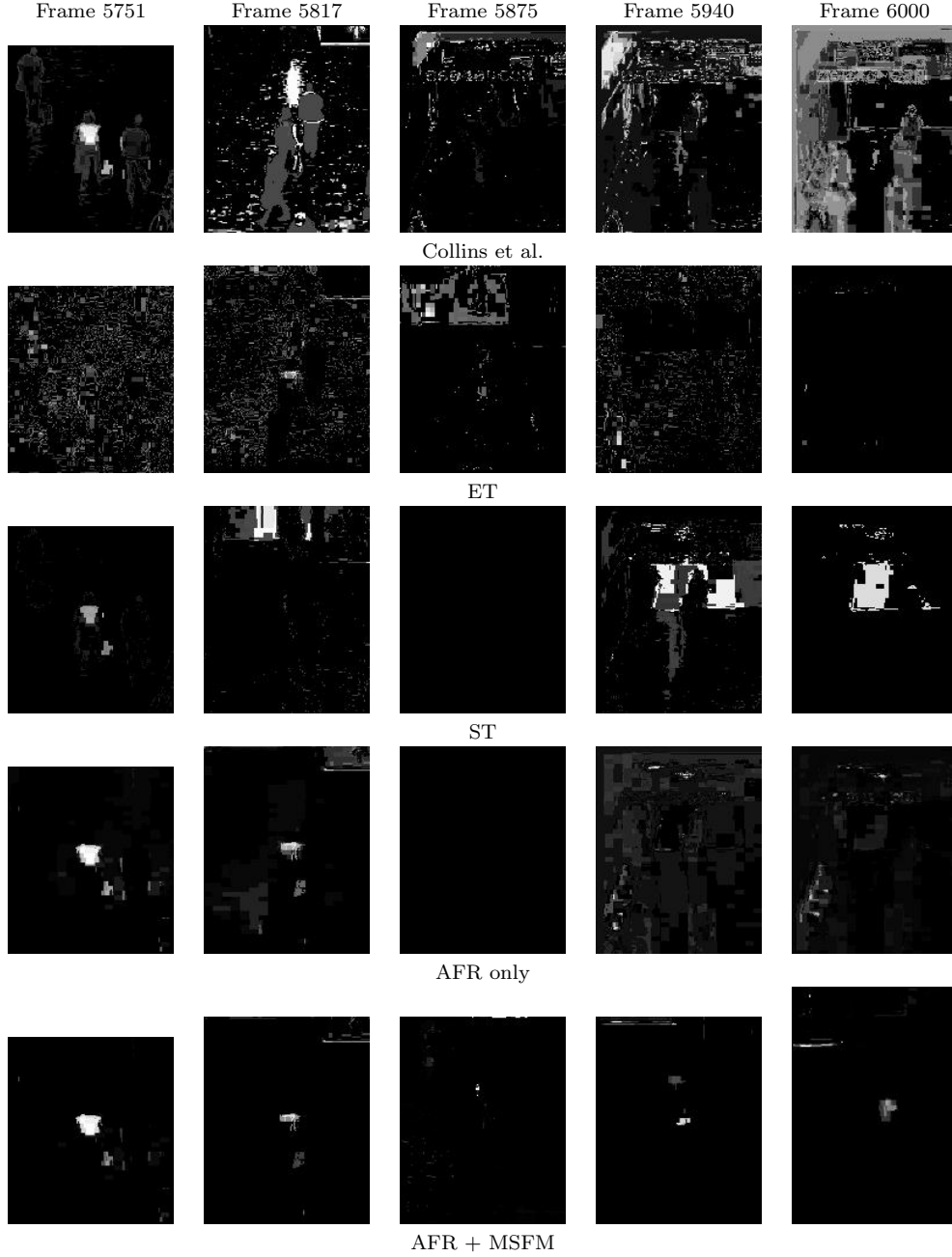


Fig. 6 Scenario B, Example 1 Confidence Maps: Severe illumination change. The centre-surround box shows tracker localisation for the object. The dashed blue box indicates manually labelled ground truth. Features selected by AFR (when locked on) resulted in improved confidence maps and pixel classification over Collins et al. and Ensemble Tracking, with emphasis on the most salient object region. AFR+MSFM improved on this further still. The SemiBoost (ST) tracker showed clean maps initially but failed earliest of all.

ground truth plot in Figure 13. As can be seen in Figure 11, AFR and AFR+MFSM consistently showed the least noisy confidence maps, with the adaptive reference models of AFR+MFSM helping to further emphasise the most salient discriminators of the target object.

Figure 12 shows statistics for the top three ranked features for this example. Again, here we see less haphazard feature ranking from AFR than Collins et

al. and Ensemble Tracking (ET), with AFR+MFSM improving on this still further. Again, the SemiBoost tracker (ST) showed the least variation, but this did not translate to accuracy of tracking in this experiment. This again suggests that the feature ranking behaviour was not as well correlated with the use of the most appropriate features, unlike for AFR and AFR+MFSM.

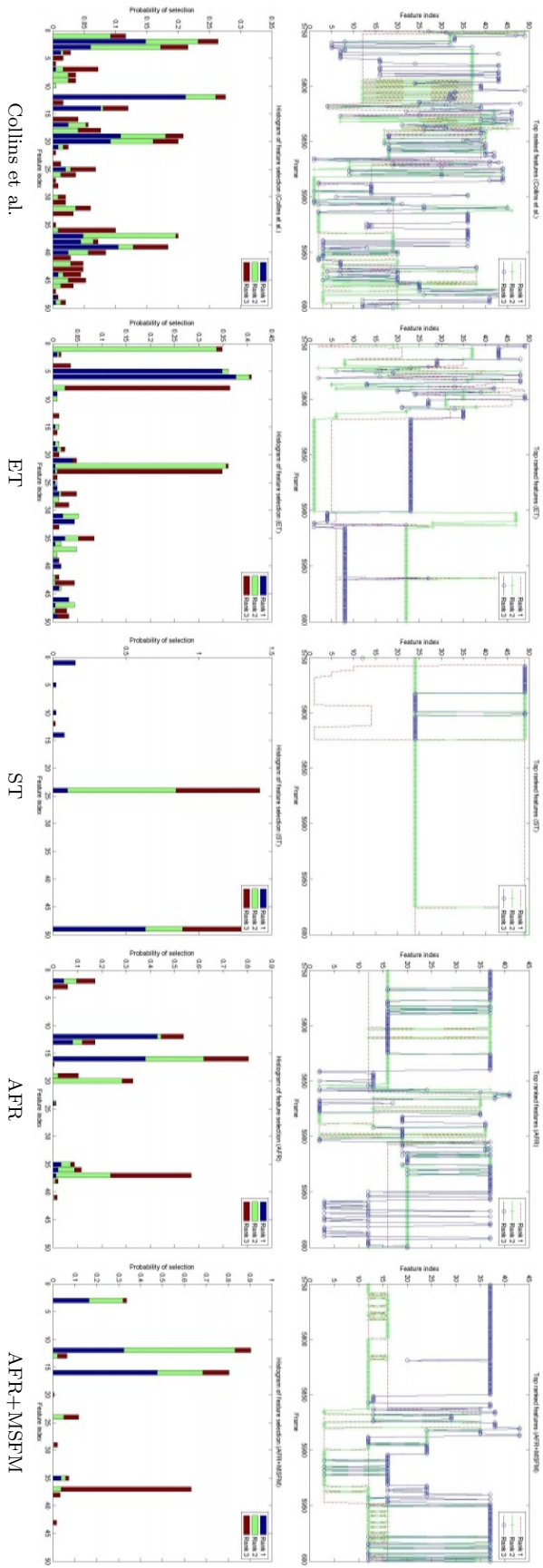


Fig. 7 Scenario B, Example 1 feature selection statistics. Top row shows top three ranked (or weighted) features for each frame and the second row the corresponding frequencies of selection for each feature in the pool. The features selected for AFR and AFR+MSFM were better discriminators and prone to fewer fluctuations than both Collins et al. and Ensemble Tracking (ET). This resulted in cleaner confidence maps (see Figure 6). The SemiBoost tracker (ST) showed least variation in feature ranking, but the tracker failed the earliest.

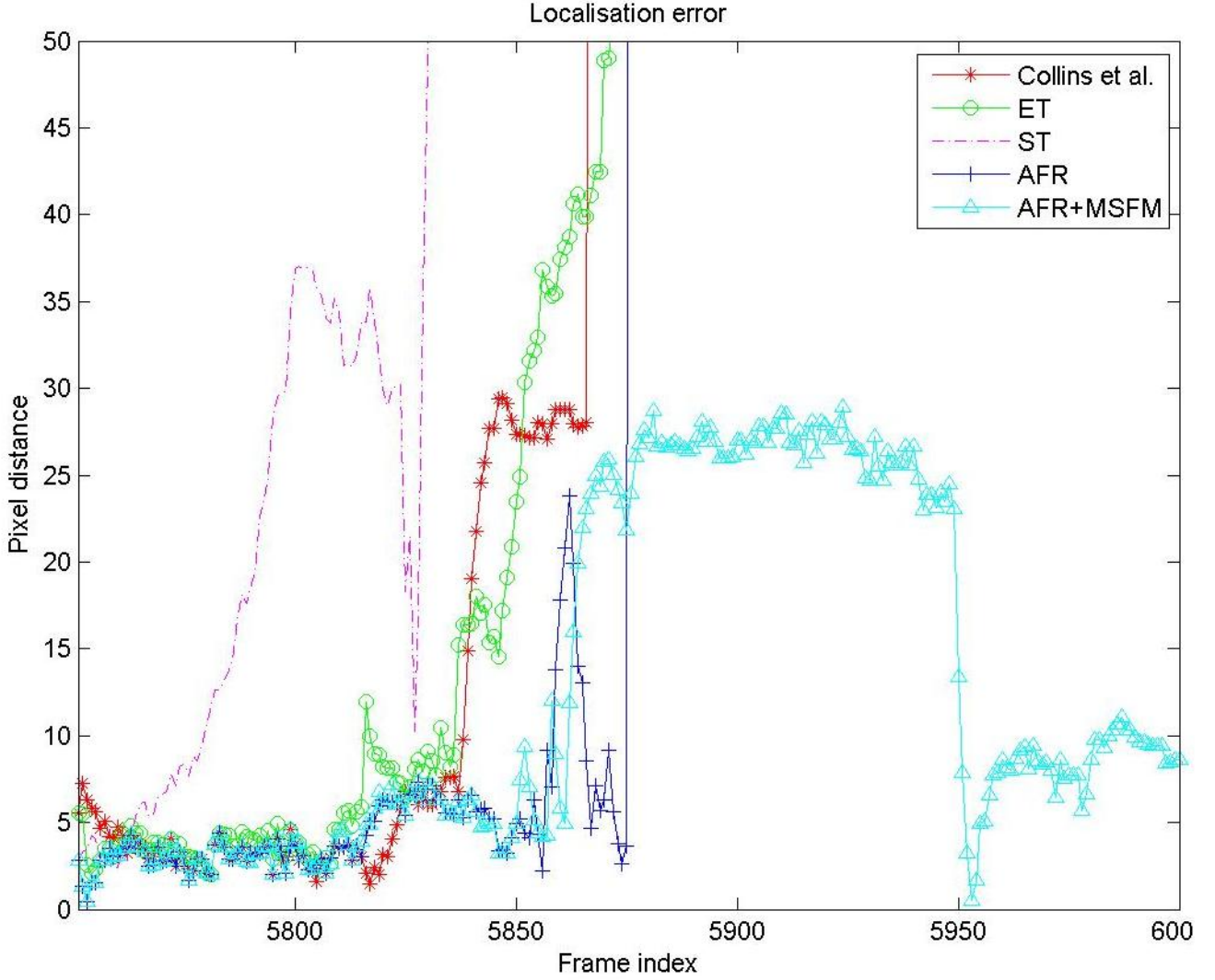


Fig. 8 Scenario B, Example 1 localisation errors in pixel distance between each tracker and manually-labelled ground truth. The SemiBoost tracker (ST) failed near the beginning of the sequence whereas Collins et al. and Ensemble Tracking (ET) were unable to cope with the darkest portion of the sequence and lost track. AFR failed shortly afterwards. However, AFR+MSFM was able to latch onto a still-salient part of the target object (a carrier bag) until sufficient brightness was restored for the tracker to reacquire the target object around frame 5950.

3.5 Scenario C: Tracking under Occlusion

In this scenario, we track a moving person at a distance from the camera in a sunny outdoor environment as they undergo severe occlusion by other moving people. This allows us to observe the ability of each tracker to deal both with temporary severe occlusion as well as sunny outdoor environments exhibiting strong shadowing effects which can lead to sudden, transient changes in appearance.

Figure 14 shows some key output frames from each tracker when applied to this sequence. Figure 15 shows the corresponding confidence maps. Specifically, Figures 14 (a)-(e) show tracking output frames and con-

fidence maps for the top ranked feature generated by Collins et al., Ensemble Tracking (ET), the SemiBoost tracker (ST), AFR and AFR+MSFM respectively. As can be verified by the ground truth plot in Figure 17, Collins et al., Ensemble Tracking and the SemiBoost tracker all failed within a few frames, whereas AFR and AFR+MSFM were able to continue tracking. Frame 49 shows the target object undergoing significant occlusion by another moving object. Shortly after, the AFR-only tracker began to get attracted to the distractor. However, AFR+MSFM remained locked onto the target object due to the up-to-date adapted feature models which were able to counteract the tendency for the frame-specific model to drift. As can be seen in

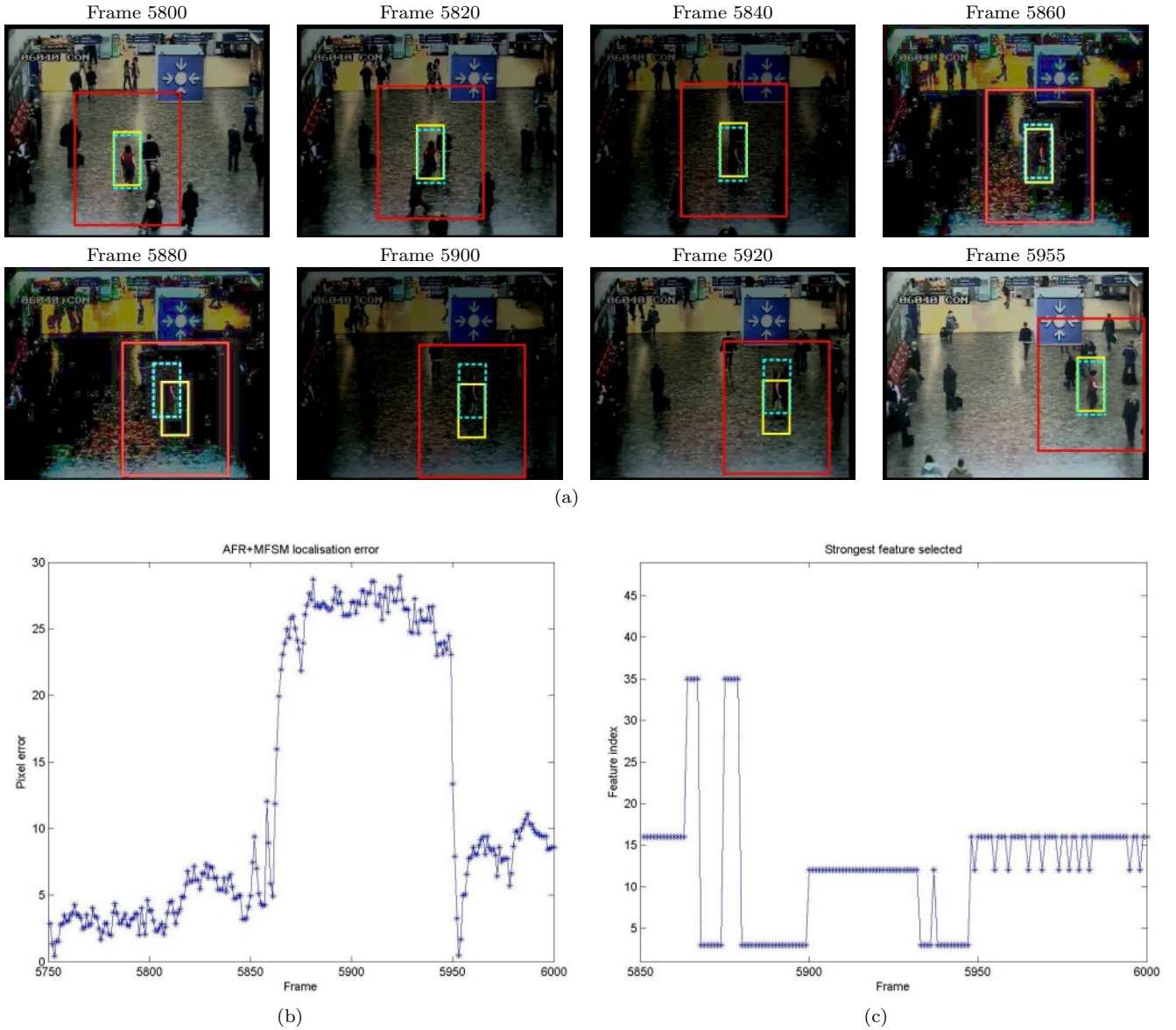


Fig. 9 Recovery of AFR+MSFM after tracking distraction due to extreme low-lighting conditions. (a) Selected frames from the sequence with ground-truth (dashed blue box) and tracker output (yellow box) superimposed. (b) Plot showing error in tracking position relative to ground-truth over the sequence. (c) Most dominant feature from frame 5850 onwards. Shortly after frame 5860 the person became difficult to distinguish due to severe low lighting conditions and the dominant feature changed from 16 to switching between 35 and 3, as can be seen from Plot (c). However, from around frame 5880, the tracker was able to attach itself to a still-salient bag being carried by the person, which was well characterised by feature 12 at that brightness range. As it started to get bright again, feature 3 reasserted itself around frame 5930 or so. Around Frame 5950 the person became visible enough for the tracker to quickly reattach itself via the previously dominant feature 16, whose reference model asserted itself to reacquire the target object. From this point the bag was also consistently emphasised via feature 12, since due to adaptive reference models its representation was strengthened during the darker period, with the result that both relevant features (16 and 12) reinforced each other in characterising the tracked person.



Fig. 10 Scenario B, Example 2 Tracker Output: Severe illumination change. The centre-surround box shows tracker localisation for the object. The dashed blue box indicates manually labelled ground truth. Ensemble Tracking (ET) and the SemiBoost tracker (ST) both failed around the frame 5850 mark, with Collins et al. becoming distracted around 100 frames later. AFR and AFR+MSFM both maintained track for the duration of the sequence. See Figure 11 for the corresponding confidence maps.

Figure 15, as for previous experiments the AFR maps were cleaner and more representative of the target when locked on, with AFR+MSFM further emphasising the most salient discriminators of target appearance and helping to maintain tracking even when AFR had failed.

The feature ranking statistics are shown in Figure 16. Although the Collins et al. statistics showed a less haphazard pattern than usual, this can be attributed to early failure and attachment to a relatively static region of the scene. The SemiBoost tracker selected the same single feature for the entire sequence due to its failure at the beginning of the sequence, latching onto a static portion of the background. AFR and AFR +

MSFM again demonstrated a better relationship between ranked features and actual performance, with the latter providing meaningful support to the former in improving actual performance.

Figure 16 shows the feature selection statistics of the top three ranked features for this outdoor scene. Again, Collins et al. and Ensemble Tracking (ET) resulted in less stable and more haphazard feature selection patterns. AFR and AFR+MSFM continued to show greater stability in feature selection. Using AFR only, the tracker was distracted after the target object underwent occlusion but the up-to-date feature models of AFR+MSFM maintained feature selection sta-

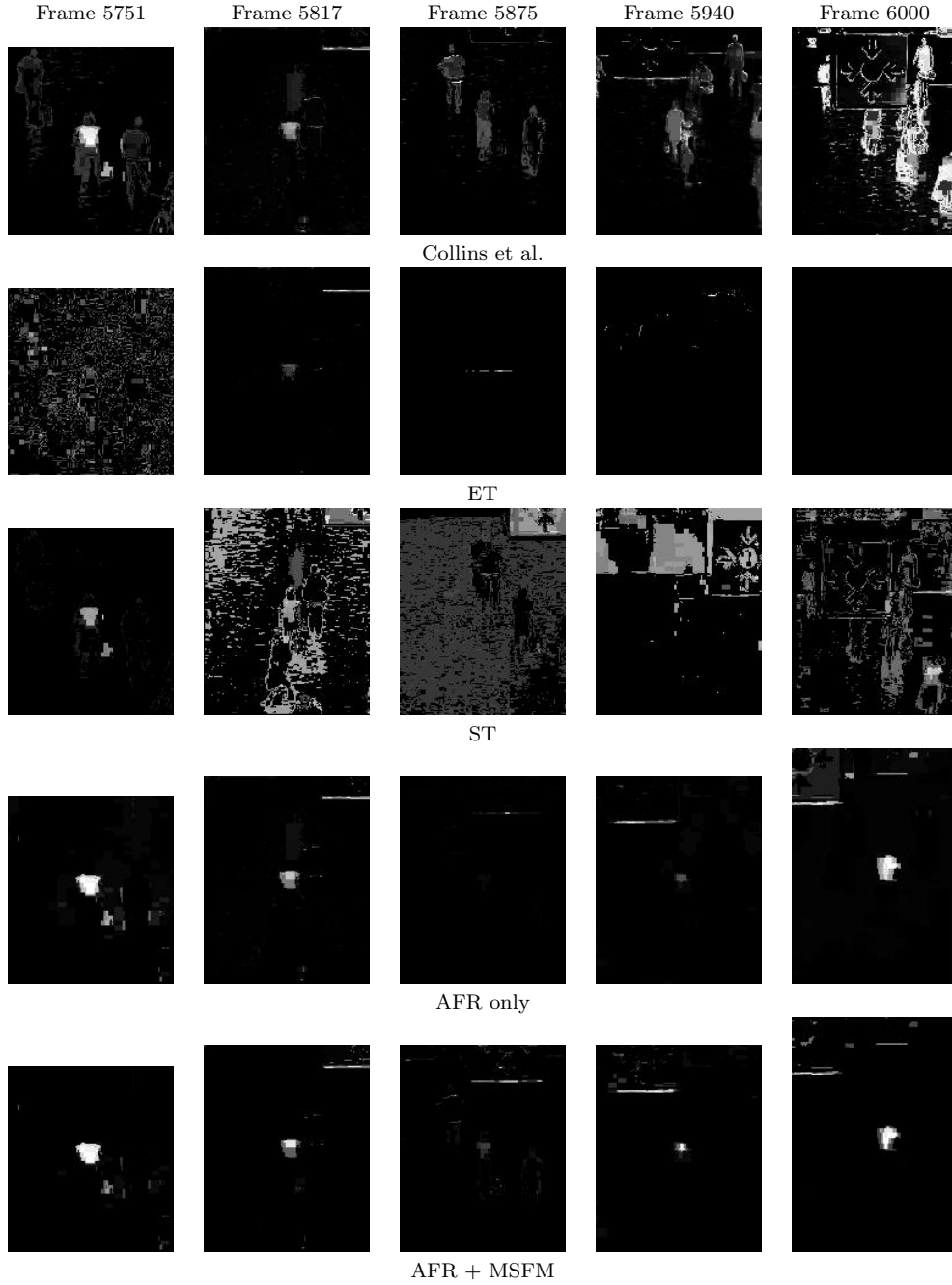


Fig. 11 Scenario B, Example 2 Confidence Maps: The AFR and AFR+MSFM trackers consistently showed the least noisy confidence maps throughout the sequence. The adaptive reference models of AFR+MSFM helped to further strengthen the most salient parts of the target object.

bility and tracking accuracy. From around Frame 52 onwards the AFR-only tracker was thrown by the occlusion as a result of pixels from the distractor polluting foreground samples and causing fluctuations in feature selection and eventual tracking failure. However, the benefit of MSFM in maintaining a relevant reference of target object appearance and offsetting the contribution of undesired pixels in ranking features was evident

by the continued stability and relevance of the features selected during occlusion.

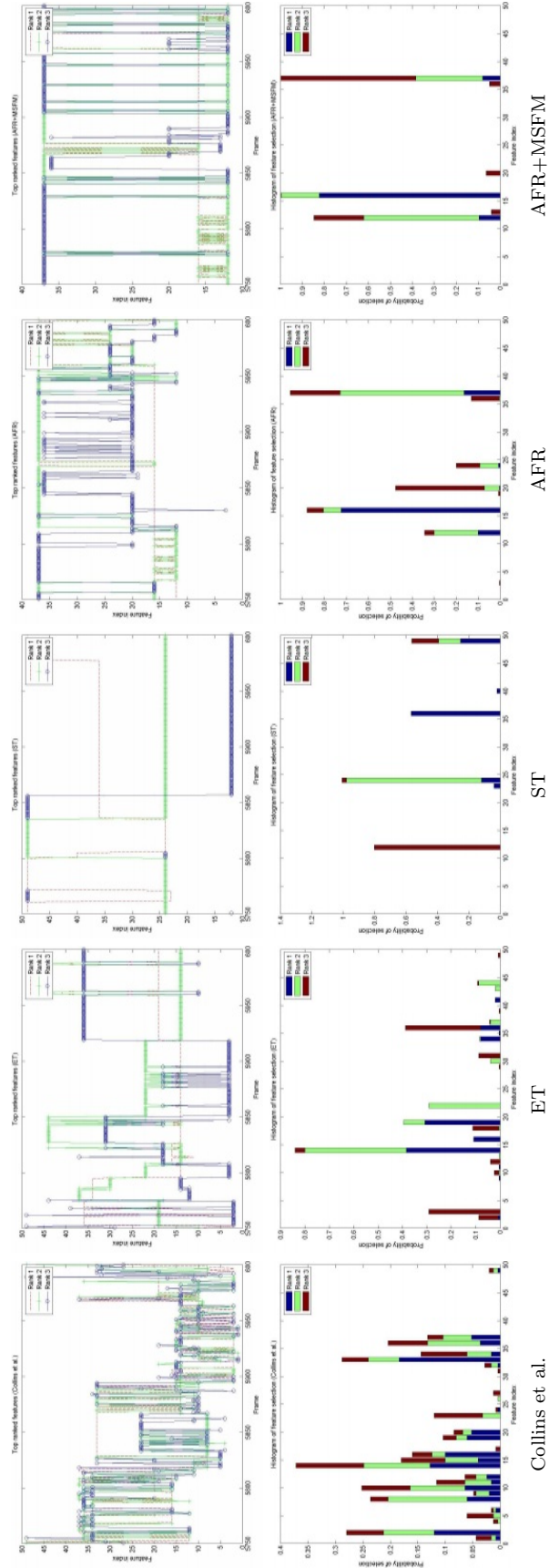


Fig. 12 Scenario B, Example 2 feature selection statistics. Top row shows top three ranked (or weighted) features for each frame and the second row the corresponding frequencies of selection for each feature in the pool. The features selected for AFR and AFR+MSFM were better discriminators and prone to fewer fluctuations than both Collins et al. and Ensemble Tracking (ET). This resulted in cleaner, more relevant confidence maps (see Figure 11). The SemiBoost tracker (ST) showed a similar rigidity as for previous experiments, but this did not translate to tracking accuracy in practice.

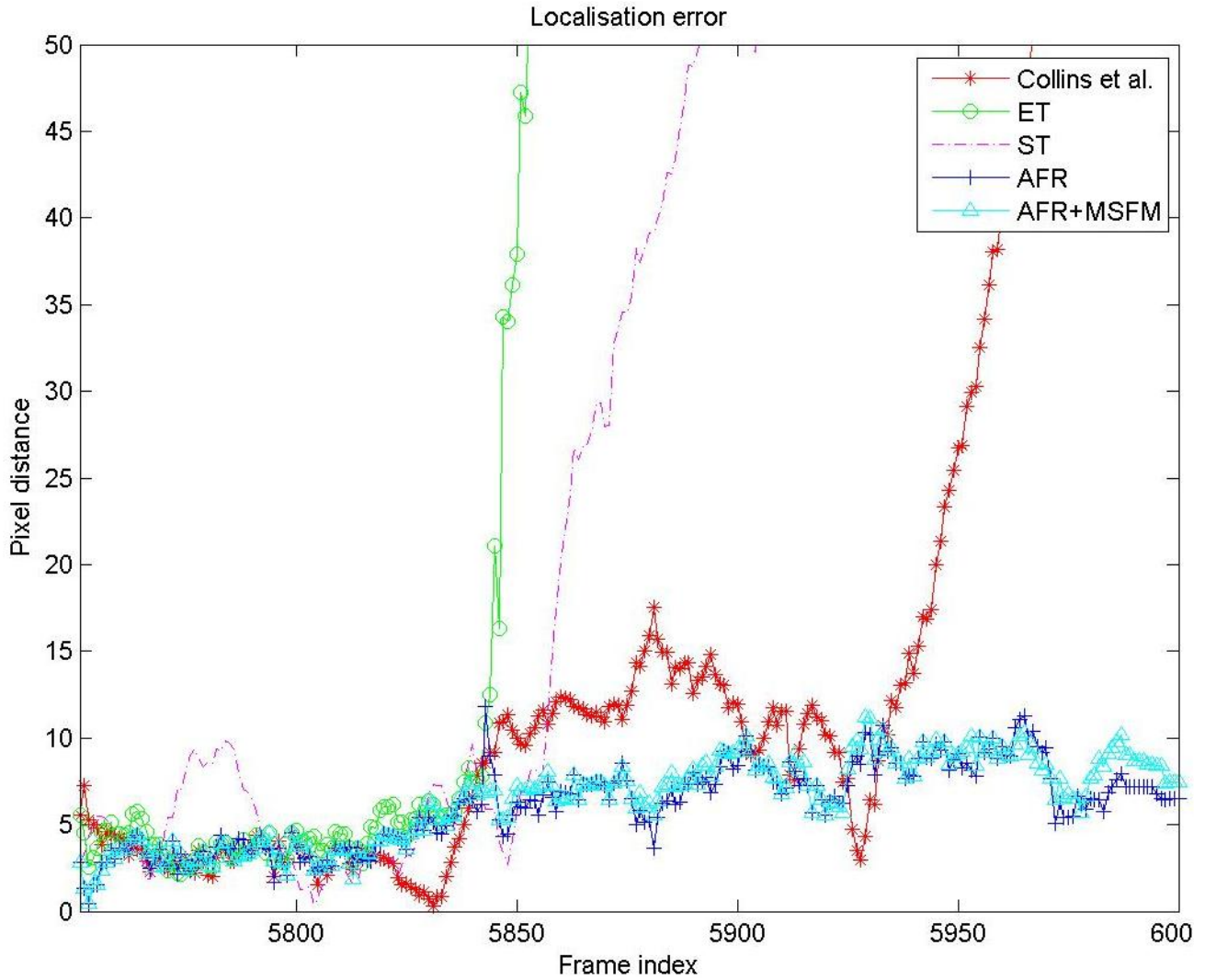


Fig. 13 Scenario B, Example 2 localisation errors in pixel distance between each tracker and manually-labelled ground truth. Ensemble Tracking (ET) and the SemiBoost tracker (ST) both failed before the halfway point of the sequence, whereas Collins et al. failed lost track around frame 5925. Both AFR and AFR+MFSM were able to maintain tracking for the duration of the sequence.

3.6 Scenario D: Tracking under Severe Occlusion and Illumination Changes

In this scenario we track a moving person at a distance from the camera in a crowded scene whilst undergoing both significant occlusion and lighting changes. Similarly to Scenario B (Section 3.4), we have two examples of changing illuminant for this scenario; the first showing a change in brightness and the second a change in colour. These changes are simulated by gradually modifying the original pixel values in sequence frames over time. Again, for brightness changes, these modifications are then gradually reversed to the original conditions.

3.6.1 Example 1 - Brightness Change

Figure 18 shows some example output frames from each tracker when applied to the sequence undergoing brightness change. Brightness was gradually reduced over a period of around 100 frames before gradually being increased at the same rate to original conditions. Specifically, Figures 18 (a)-(e) show tracking outputs for Collins et al., Ensemble Tracking (ET), the SemiBoost tracker (ST), AFR and AFR+MSFM respectively. Figure 19 shows the corresponding confidence maps. As can be verified from the ground truth plot in Figure 21, Collins et al. and Ensemble Tracking became distracted around the same time, shortly after frame 90 when the target object was both heavily occluded and lacking in saliency due to a significant lack of brightness. AFR



Fig. 14 Scenario C Tracker Output: Tracking under occlusion. The dashed dark blue box shows ground truth. Collins et al., Ensemble Tracking (ET) and the SemiBoost tracker (ST) failed within a few frames, whereas AFR and AFR+MSFM continued to track successfully. AFR began to fail around Frame 52 due to occlusion (see Figure 17) but AFR+MSFM resisted distraction due to up-to-date feature reference models.

and AFR+MSFM were both able to successfully hold the target for the duration of the sequence. Their corresponding confidence maps in Figure 19 again show the familiar relationship of relatively clean maps from AFR with the addition of MSFM providing further improvement by way of emphasising the most salient feature values. The supportive nature of MSFM for AFR was also reflected in the feature selection statistics of Figure 20 where the more controlled pattern of feature selection correlated with the quality of the resulting confidence maps and actual tracker performance.

In this experiment, the SemiBoost tracker was also able to successfully track the target object for the duration and as before showed significant rigidity in fea-

ture ranking and selection (with the same feature being ranked highest throughout). The resulting confidence maps were the cleanest of all. However, given previous results, this is more likely to be through fortune rather than any structural relationship between the feature ranking/selection approach and the most relevant features for the sequence in particular.

3.6.2 Example 2 - Colour Change

Figure 22 shows some key output frames from each tracker when applied to the sequence undergoing colour change. The red channel was gradually reduced and the blue channel gradually increased over a period of around 100 frames. Specifically, Figures 22 (a)-(e) show

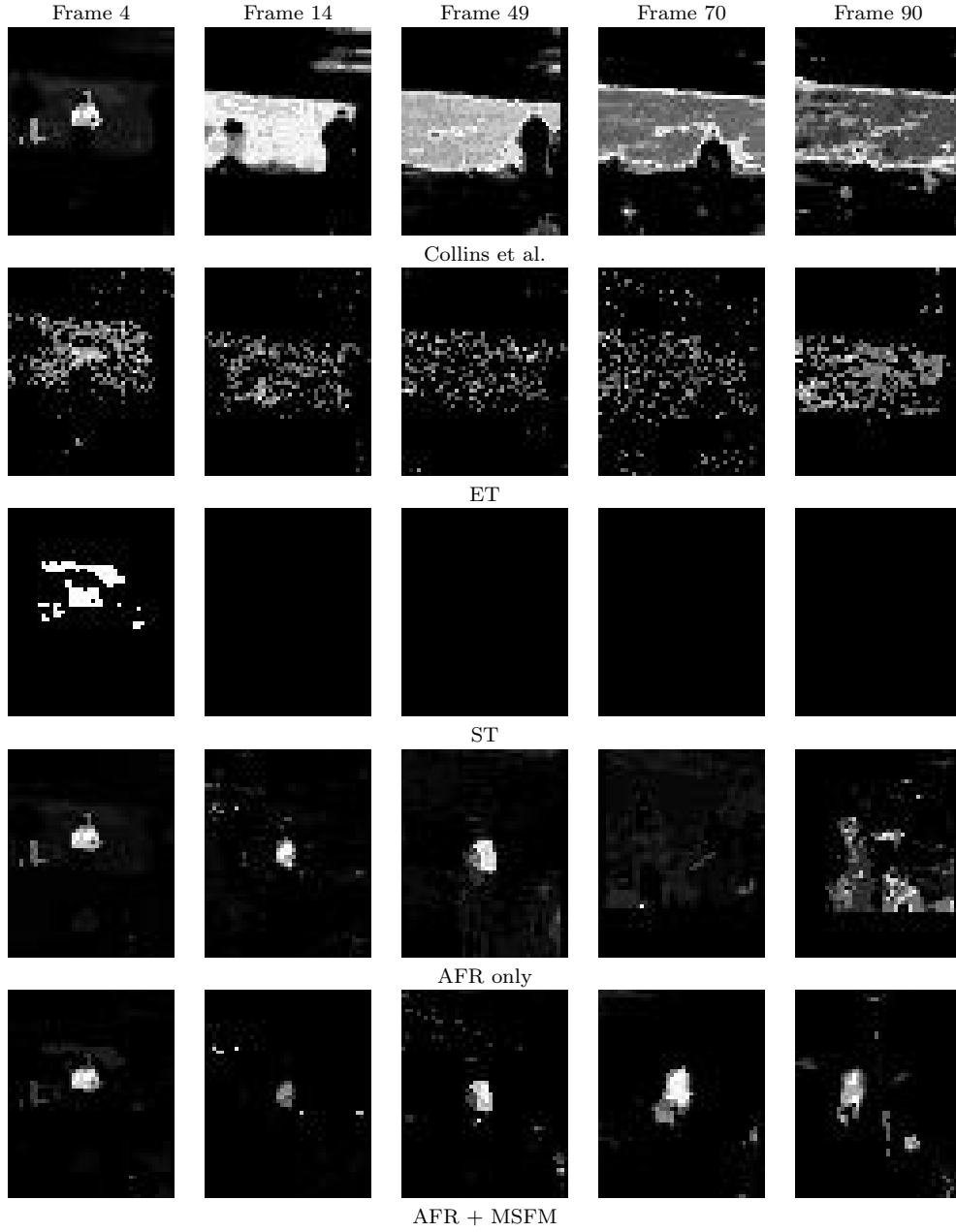


Fig. 15 Scenario C Confidence Maps: The top three rows represent trackers that failed early and so expectedly depict noisy maps bearing no relevance to the tracking target. AFR maps were clean and representative up until failure, whereas AFR+MSFM again improved on AFR by further reducing noise and consequently improving actual tracking performance.

tracking outputs for Collins et al., Ensemble Tracking (ET), the SemiBoost tracker (ST), AFR and AFR + MSFM respectively. Figure 23 shows the corresponding confidence maps.

As can be verified from the ground truth plot in Figure 25, this time the SemiBoost tracker failed shortly after frame 10, losing the target completely. The other four trackers continued up to around frame 95, where the Avidan tracker suddenly lost the target object. The Collins et al. and AFR trackers also became attracted

to a nearby distractor partially occluding the target object; however, the AFR tracker was able to recover within around 15 frames. The Collins et al. tracker also recovered around frame 120. This was likely helped by the fact that the target object stopped moving, although surrounding distractors continued to move and cast shadows on the target object and the lighting continued to change. AFR+MSFM successfully maintained a lock on the object for the duration of the sequence.

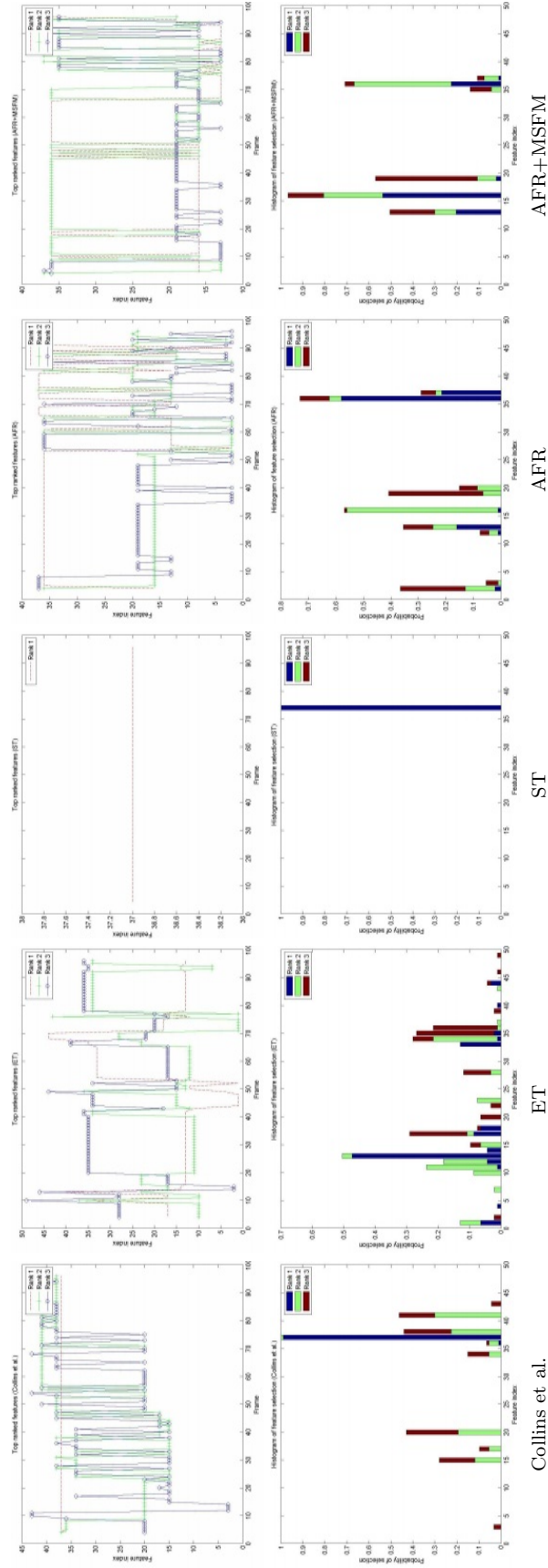


Fig. 16 Scenario C feature selection statistics. Top row shows top three ranked (or weighted) features for each frame and the second row the corresponding frequencies of selection for each feature in the pool. The features selected for AFR and AFR+MSFM were better discriminators and prone to fewer fluctuations than Ensemble Tracking (ET), with the AFR+MSFM again further improving on AFR alone. This resulted in cleaner confidence maps (see Figure 15). Collins et al. showed less fluctuation than normal, but this is attributable to its attachment to a relatively static portion of the background following tracking failure. The SemiBoost tracker (ST) selected the same single feature for the whole sequence, but again this was largely due to tracking failure occurring at the beginning of the sequence.

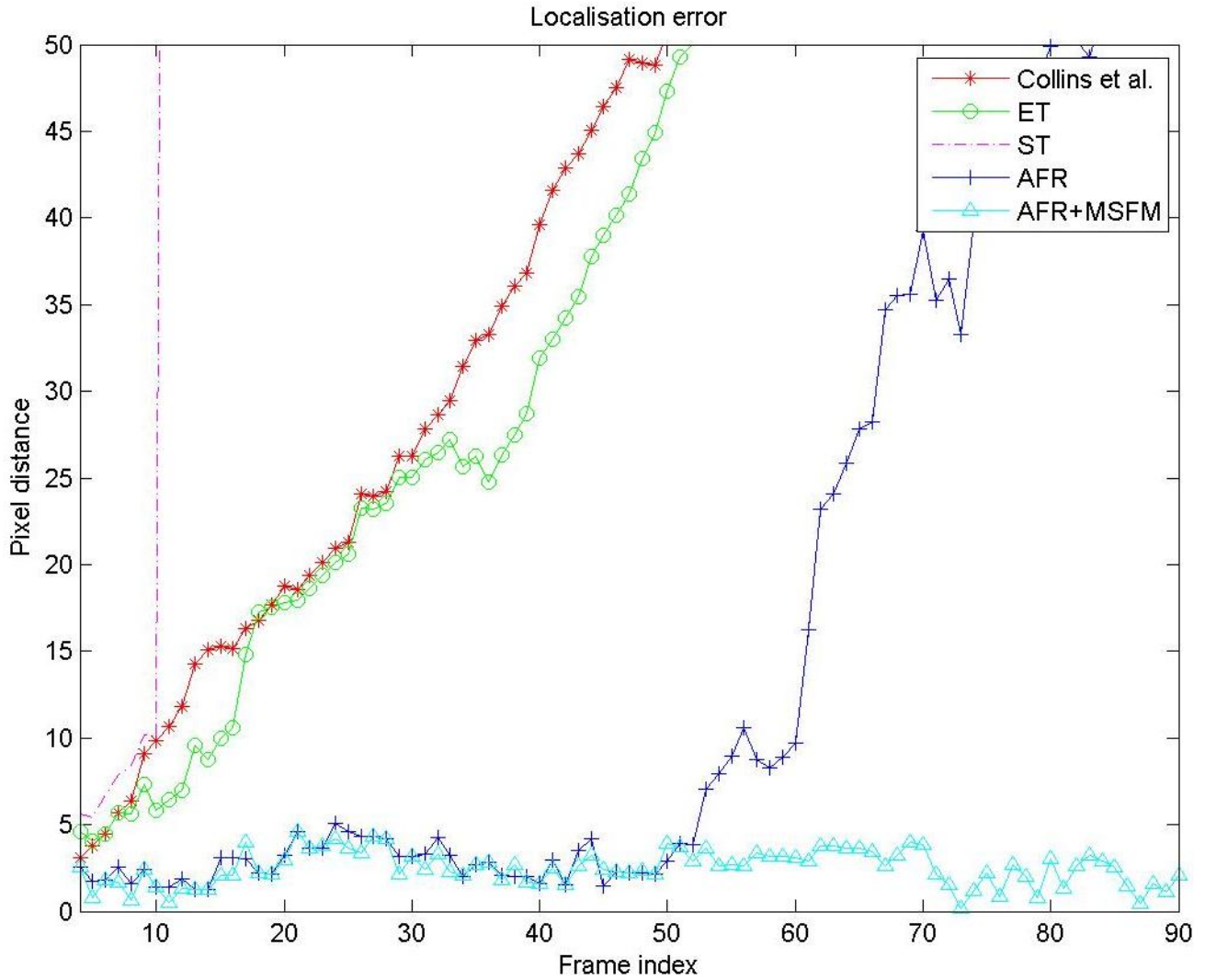


Fig. 17 Scenario C localisation errors in pixel distance between each tracker and manually-labelled ground truth. Collins et al., Ensemble Tracking (ET) and the SemiBoost tracker (ST) failed almost instantly, whereas AFR was able to track the target before being distracted by a significant moving occluder shortly after frame 50. AFR+MFSM maintained a lock on the target for the duration of the sequence.

The confidence maps in Figure 23 were relatively clean for all trackers when they were locked on, with failed frames showing noisy confidence maps indicative of the models' poor correlation with the image data given the bounding box position. The AFR+MFSM combination again provided the balance to maintain tracking via a more appropriate selective adaptation of reference models.

4 Conclusions

Feature selection has more commonly been used in recent years as part of a general classification-based approach to robust object tracking. Previous methods have

not considered the main issue of redundancy in feature selection. They have also not addressed the problem of balancing the necessary flexibility required for models to keep track of changing long-term object appearance against the need to prevent the common problem of model drift. In this paper, we proposed accordingly a general feature-selection based tracking framework for robust tracking in difficult conditions where objects undergo significant long-term change. This involves combining the knowledge of trends in target object appearance over time with appearance characteristics in each frame to help overcome difficulties such as lighting change and partial occlusions. The benefit of this approach over previous methods (such as Collins et al

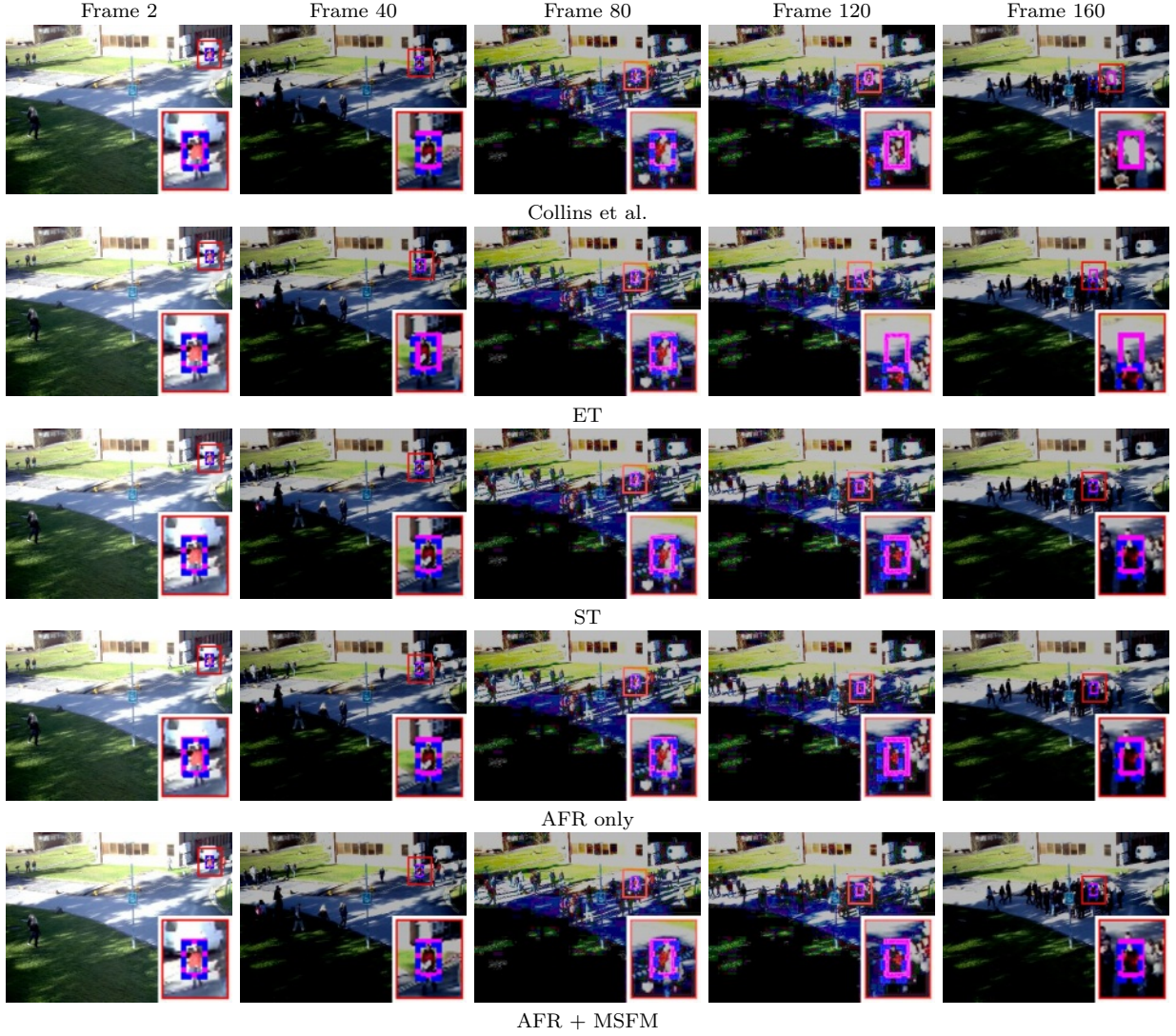


Fig. 18 Scenario D, Example 1 Tracker Output: Tracking under occlusion and lighting change. Frames 80 and 120 were gamma corrected for reader clarity. The dashed dark blue box shows ground truth. Collins et al. and Ensemble Tracking (ET) failed around frame 90, whereas the SemiBoost tracker (ST), AFR and AFR+MSFM all continued to maintain a lock successfully for the duration of the sequence.

(2005); Avidan (2007); Liang et al (2007); Grabner et al (2006, 2008)) are twofold:

1. Features are selected according to a measure of their correlation with other features in the pool in order to reduce redundancy among those selected at each frame;
2. Feature reference models are *dynamic* rather than static and bypass unrealistic assumptions made by previous methods, in the process contributing significantly to the overall strategy of trading off short-term evidence with long-term trends.

In implementing this strategy, we formulated a novel tracking framework called Adaptive Multi-feature Association (AMA) which comprises two methods for im-

proving dynamic feature ranking and selection in tracking:

1. Attribute-based Feature Ranking (AFR) which combines two attribute measures, (a) a measure of discriminability and (b) a measure of independence to other features;
2. Multiple Selectively-adaptive Feature Models (MSFM) which involves maintaining a dynamic feature reference of target object appearance. These are updated selectively against current image evidence for ranking features and classifying pixels in each frame.

We compared the performance of our framework to the established methods of Collins et al (2005), Avidan (2007) and Grabner et al (2008) using four chal-

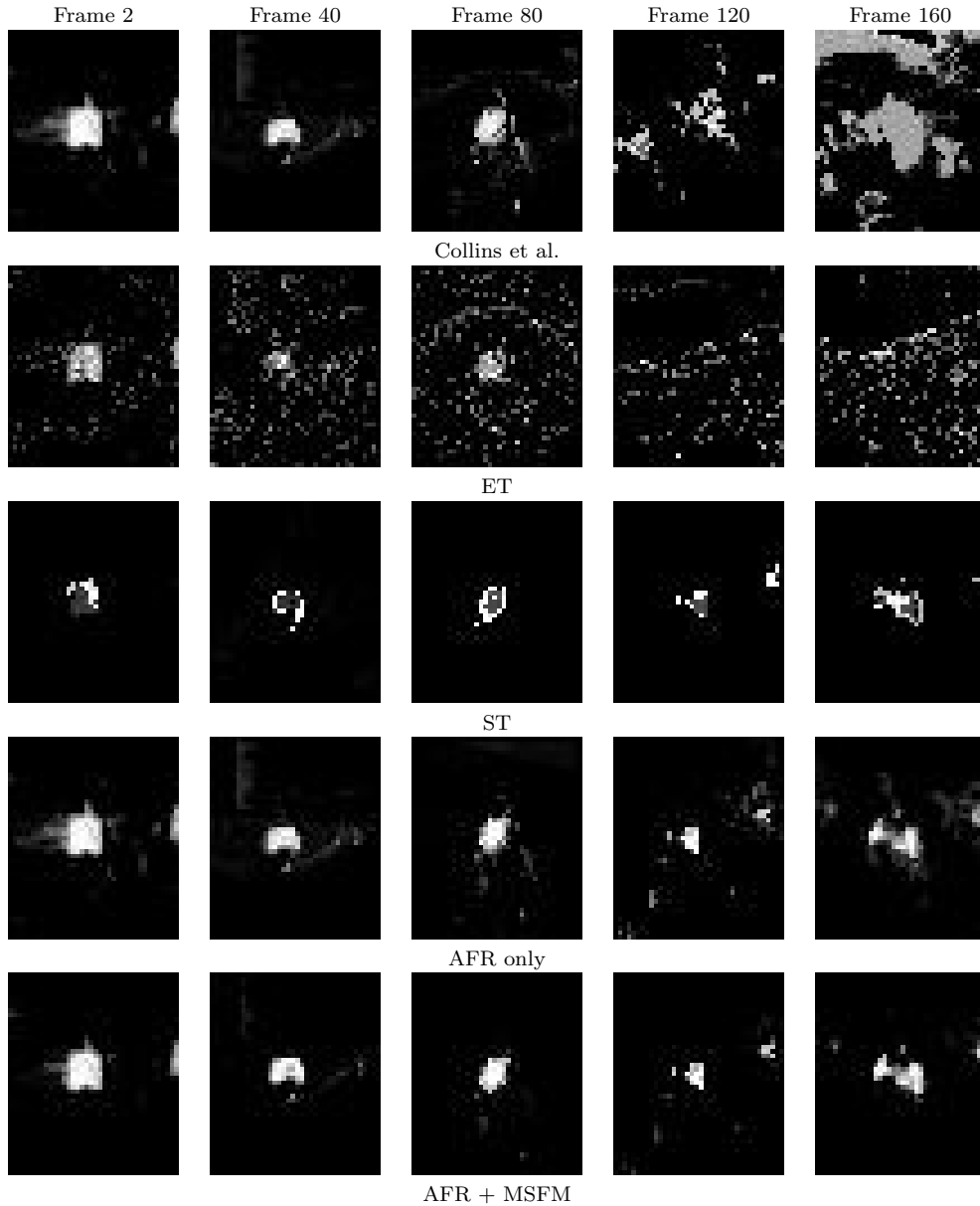


Fig. 19 Scenario D, Example 1 Confidence Maps: The SemiBoost tracker (ST), AFR and AFR+MSFM all showed the most consistently representative confidence maps, with MSFM again showing its ability to support AFR in improving pixel classification. The SemiBoost tracker showed the cleanest maps as a result of its lack of flexibility in feature ranking (see Figure 20).

lenging real-world scenarios. We showed in Scenario A that AFR selected more stable features which led to cleaner confidence maps that emphasised the most salient discriminative part of the tracking target object. Using AFR+MSFM improved pixel classification due to the up-to-date feature reference models and further improved the confidence maps. In Scenario B, we showed two examples of how these factors helped to overcome severe lighting changes which caused the Collins et al., Ensemble Tracking and SemiBoost trackers to fail. Specifically, we demonstrated how the dynamic refer-

ence models enabled the tracker to adapt and recover in the face of a temporary and significant lack of saliency of the tracked object due to a severe lighting deficiency in Example 1 as well as preventing failure during a significant ambient colour change in Example 2. In Scenario C, we tested robustness under significant partial occlusion in a cluttered environment. AFR tracked longer than Collins et al., Ensemble Tracking or SemiBoost tracking but was perturbed by occlusion. We showed how AFR+MSFM overcame this through the use of MSFM to maintain the relevance of feature refer-

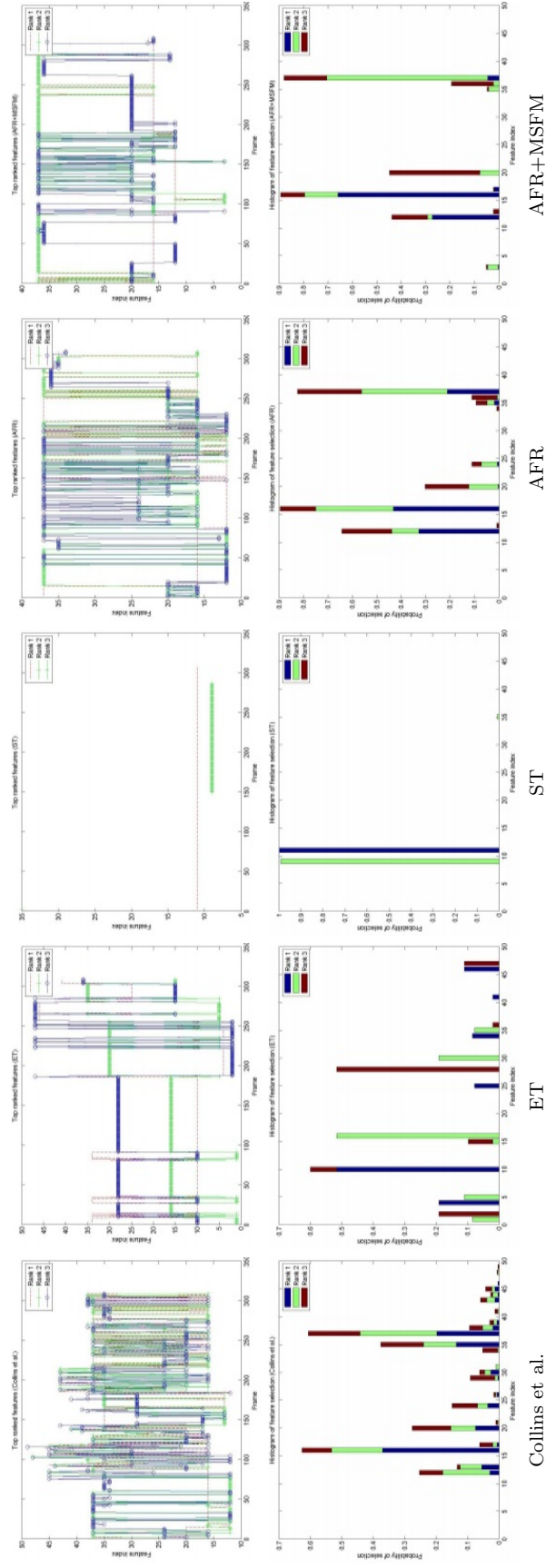


Fig. 20 Scenario D, Example 1 feature selection statistics. Top row shows top three ranked (or weighted) features for each frame and the second row the corresponding frequencies of selection for each feature in the pool. The features selected for AFR and AFR+MSFM were better discriminators and prone to fewer fluctuations than both Collins et al. and Ensemble Tracking (ET). This resulted in cleaner confidence maps (see Figure 19), with AFR+MSFM again demonstrating meaningful improvement over AFR alone. The SemiBoost tracker (ST) again showed significant rigidity in feature ranking, which in this example proved adequate for tracking to succeed.

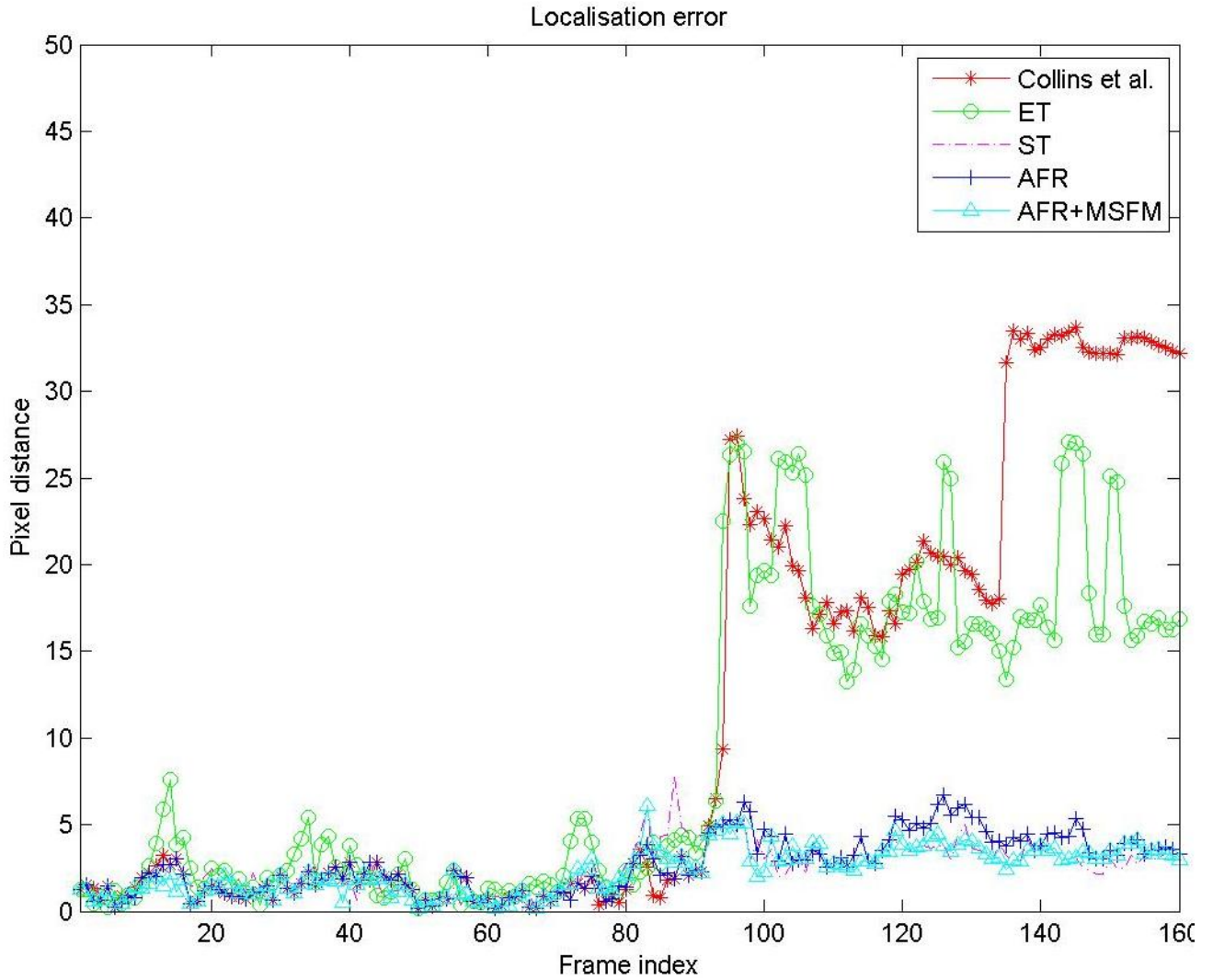


Fig. 21 Scenario D, Example 1 localisation errors in pixel distance between each tracker and manually-labelled ground truth. Collins et al. and Ensemble Tracking (ET) were both distracted around frame 90, whereas the SemiBoost tracker (ST), AFR and AFR+MSFM all succeeded in tracking throughout the sequence.

ence models and counteract the tendency of the frame-specific models to drift terminally under such conditions. Consequently, the target object was successfully tracked subsequent to reappearance from occlusion. In Scenario D we examined tracking under both difficulties of changing lighting and clutter/occlusion simultaneously, again with two examples; one of severe brightness change and the other of strong colour change. For the former, the SemiBoost, AFR and AFR+MSFM trackers were successful, whereas in the latter, only AFR + MSFM tracked throughout.

In these experiments, AFR was able to select more descriptive, appropriate features more consistently than Collins et al., Ensemble Tracking and SemiBoost tracking by virtue of the CCA-based Independence attribute. MSFM also consistently showed in every experiment

that it was able to strongly support AFR in improving pixel classification by way of maintaining updated models which significantly reduced the chance of model drift, resulting in greater robustness in tracking. This suggests a more meaningful balance between flexibility and rigidity to obtain a better level of practical real-world consistency. Furthermore, the AMA framework is less complex than both Ensemble Tracking and SemiBoost tracking.

We aim to conduct further experiments comprising larger feature pools incorporating geometric statistical features such as oriented gradients (Dalal and Triggs (2005)) as well as alternative colour spaces. Also, currently this framework focuses on the tracking of a single object. A natural way forward is to integrate data association techniques such as Joint Probabilistic Data As-

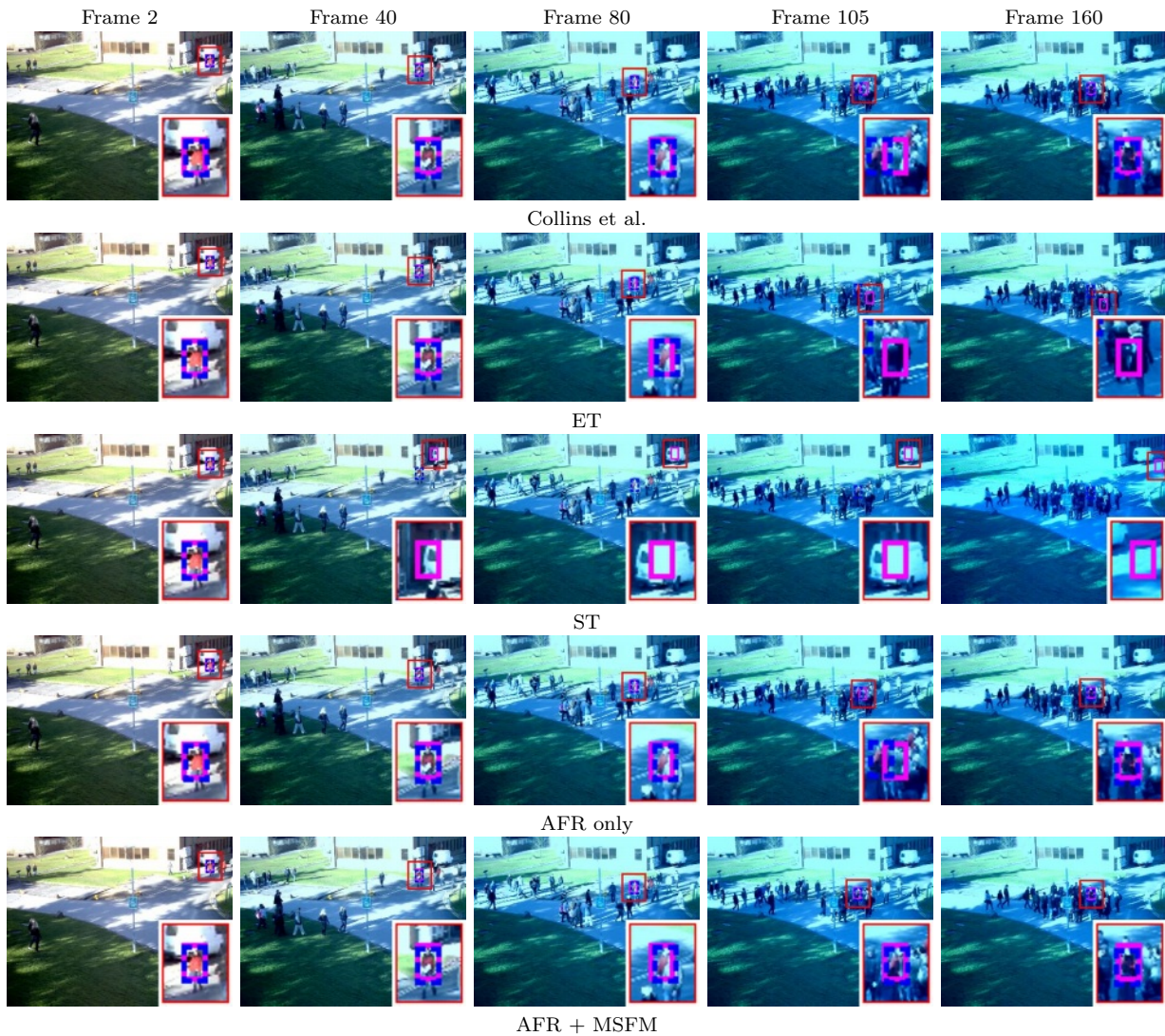


Fig. 22 Scenario D, Example 2 Tracker Output: Tracking under occlusion and lighting change. The dashed dark blue box shows ground truth. The SemiBoost tracker (ST) failed within ten frames while the others continued to track. Collins et al., Ensemble Tracking (ET) and AFR all became distracted around frame 95, with AFR recovering within fifteen frames and Collins et al. recovering within thirty frames. AFR+MSFM stayed locked onto the target for the duration of the sequence. See Figure 25 for the ground truth plot.

sociation Filters (JPDAFs) (Fortmann et al (1983)) to handle multiple tracked objects simultaneously in difficult scenarios and in a unified and rigorous manner.

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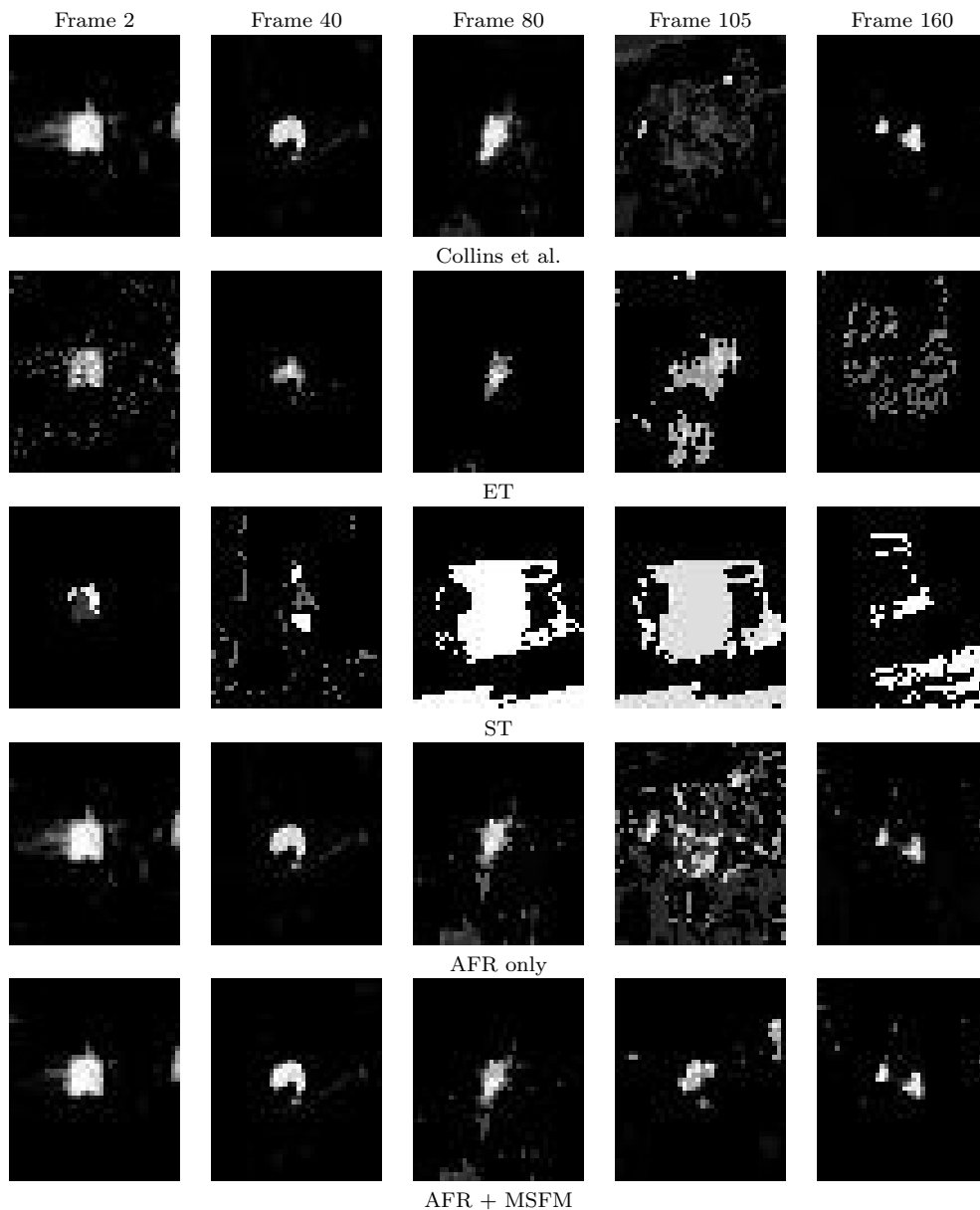


Fig. 23 Scenario D, Example 2 Confidence Maps: Due to early failure, the SemiBoost tracker (ST) showed the least relevant confidence maps overall, with Ensemble Tracking (ET) also noisy following an unrecoverable failure around frame 95. As has been consistently demonstrated, MSFM helped to support AFR by reducing noise and improving tracking performance. Collins et al. showed a very similar quality in its confidence maps for this sequence although tracking performance was not up to the consistency of AFR and AFR+MSFM.

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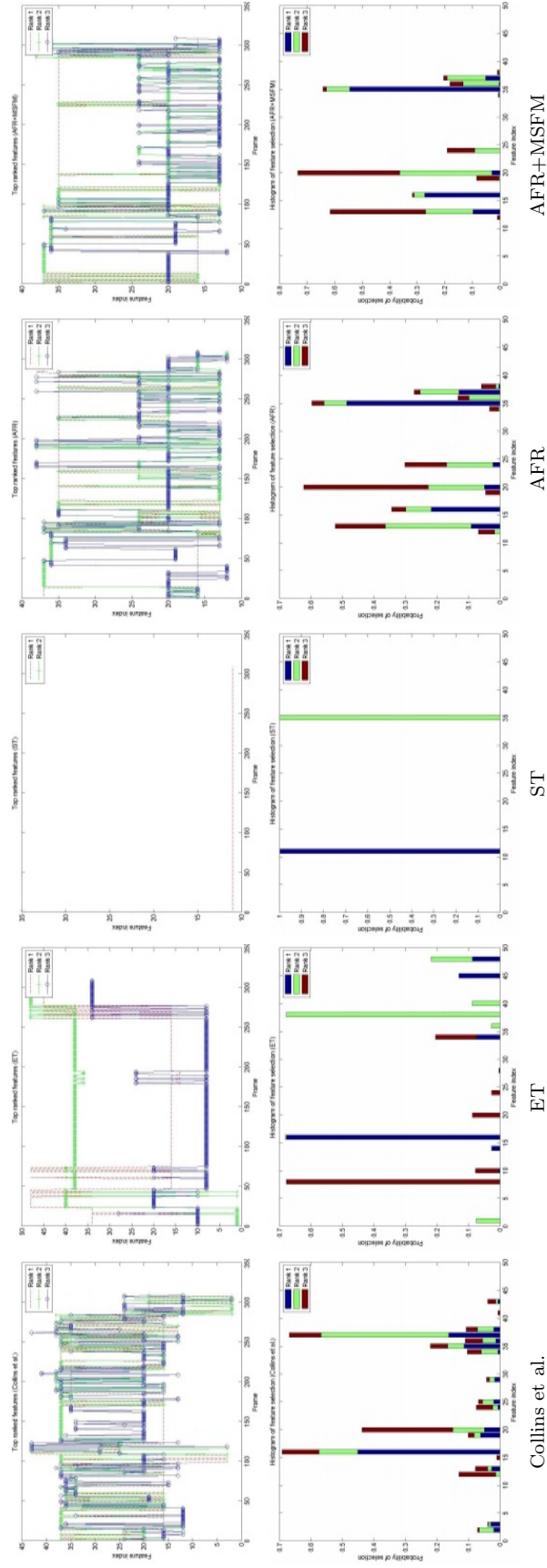


Fig. 24 Scenario D, Example 2 feature selection statistics. Top row shows top three ranked (or weighted) features for each frame and the second row the corresponding frequencies of selection for each feature in the pool. As for all previous experiments, AFR+MSFM added further control to the feature ranking and selection statistics of AFR alone which in turn translated into actual performance (see Figure 25). The SemiBoost tracker (ST) again showed significant rigidity in feature ranking but failed early on. The statistics for Collins et al. and Ensemble Tracking (ET) showed a less haphazard characteristic than normal, but neither were able to track as consistently as AFR or AFR+MSFM.

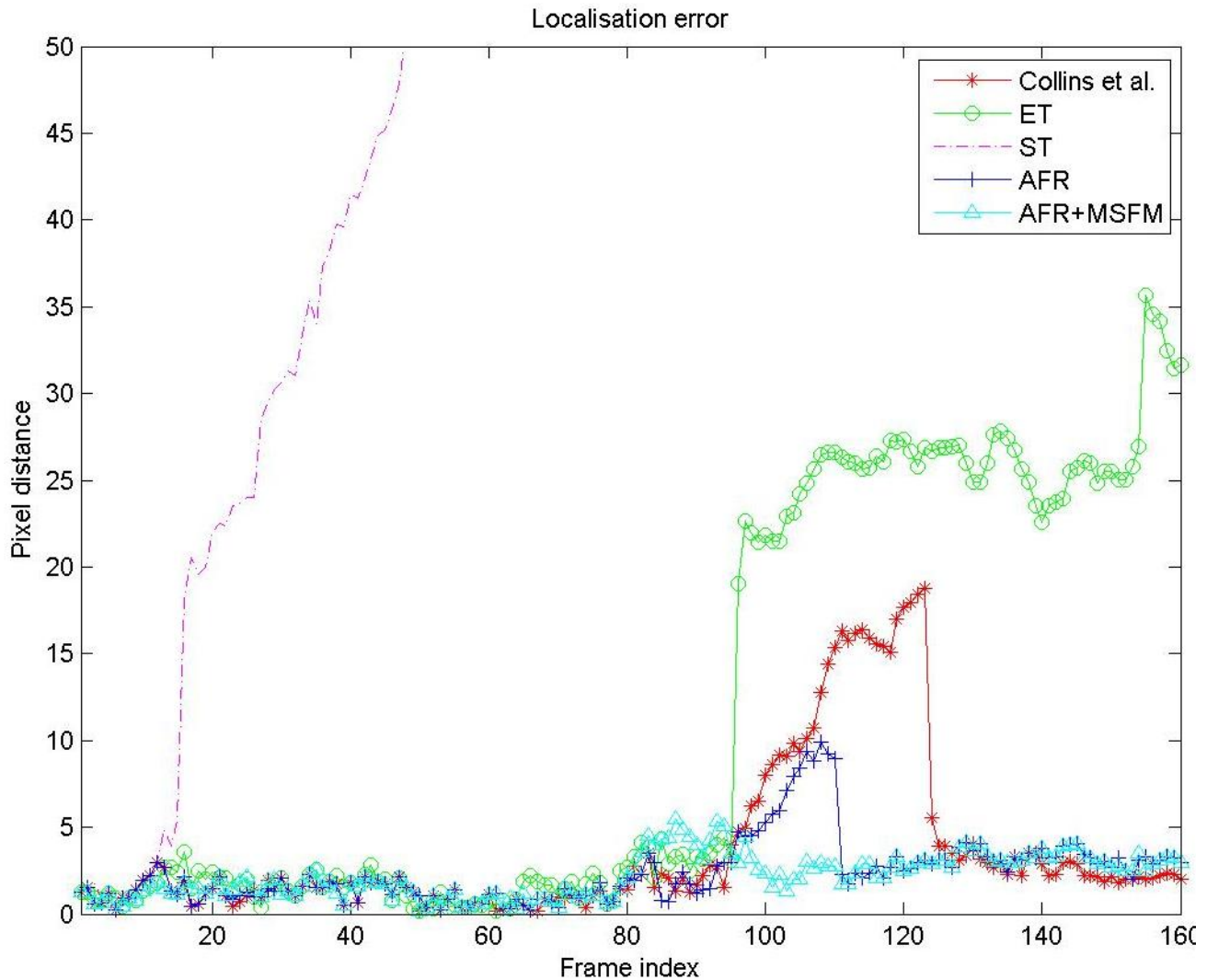


Fig. 25 Scenario D, Example 2 localisation errors in pixel distance between each tracker and manually-labelled ground truth. The SemiBoost tracker (ST) fails very early, whereas Ensemble Tracking (ET) continues up to around frame 95 where it fails irreversibly. Both Collins et al. and AFR become distracted by the same proximal distractor around the same time, with the latter recovering sooner than the former. Only AFR+MSFM was able to maintain a lock for the duration of the sequence.

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